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# Localization of nuclear wave functions of lithium in $[\text{Li}^+\text{@C}_{60}]\text{PF}_6^-$ : molecular insights into two-site disorder–order transition†

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Lithium endohedral fullerene,  $\text{Li}^+\text{@C}_{60}$ , is a porous system ideal for studying the quantized translational motion of the  $\text{Li}^+$  nucleus under subnanoscale confinement. The quantized nuclear motion strongly depends on the anharmonic and polarizable adsorbent potential within the  $\text{C}_{60}$  cage, which can be perturbed by cage distortion and/or exterior ions. In our recent paper, H. Ando and Y. Nakao, *Phys. Chem. Chem. Phys.*, 2021, **23**, 9785–9803, we focused on a  $[\text{Li}^+\text{@C}_{60}]\text{PF}_6^-$  salt and theoretically investigated how the  $\text{Li}^+$  ion in each  $\text{C}_{60}$  cage is simultaneously localized at two equivalent disordered sites in  $24\text{ K} < T \ll 100\text{ K}$ . At 24 K, the salt exhibits a disorder–order transition, whereby every  $\text{Li}^+$  ion becomes mostly localized at one of the two disordered sites below that temperature. Herein we discuss the origin of this transition with special attention to the local structural distortion and intermolecular interactions. Using the Fourier grid Hamiltonian method and a model function that fits a post-Hartree–Fock potential energy surface, we obtained hundreds of low-energy nuclear wave functions of  $\text{Li}^+$  confined within the cage. The weak distortions of the  $\text{C}_{60}$  cage and the  $\text{PF}_6^-$  coordination sphere below 24 K and concurrent inversion-symmetry breaking affect intermolecular interactions, thus making the wave functions of the nuclear ground state and several low-energy excited states localized around the experimental high-occupancy disordered site. This demonstrates that the distortions correlate closely with the two-site disorder–order transition. Finally, we reveal that two absorption peaks in the terahertz frequency range show substantial blueshifts upon cooling below 24 K, which serve as fingerprints of the transition.

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## 1. Introduction

Nanoporous materials, such as porous carbons,<sup>1,2</sup> covalent organic frameworks,<sup>3</sup> and metal–organic frameworks,<sup>4–6</sup> contain innumerable subnanoscale pores, which provide accommodations for guest molecules or ions. The controllable porosity and the host framework's designability enable us to enrich the conceptual understanding of host–guest science and to explore unique properties (*e.g.*, guest storage,<sup>1,4–7</sup> separation,<sup>2,5</sup> sensing,<sup>8,9</sup> and energy conversion<sup>10,11</sup>). When a lightweight guest species is trapped in a subnanopore, its translational nuclear motion, as well as its rotational and vibrational motions, is often quantized and different from that in bulk matter.<sup>12–24</sup> The quantum nature of electrons also plays a key role in the coupled behavior of a host and guests.

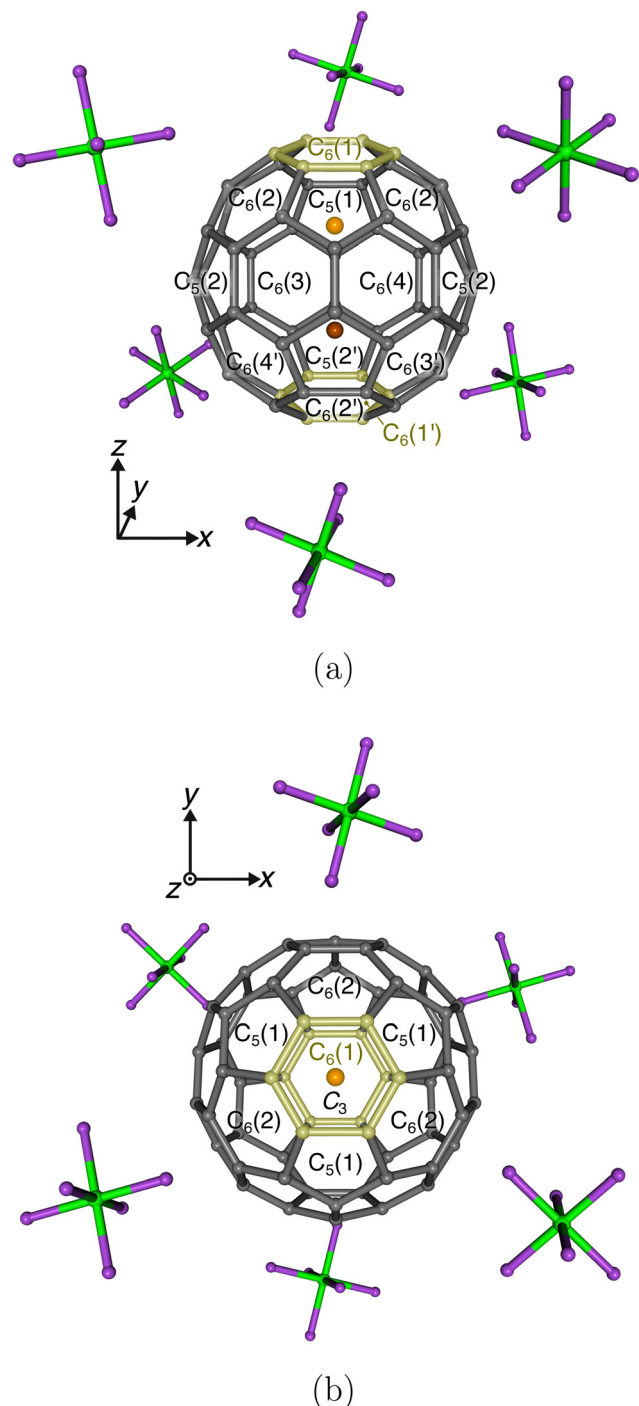
Specifically, the electronic state of the host framework can flexibly respond to the guest species migrating from one interstitial site to another, thereby, for instance, inducing electronic polarization and dispersion interactions.<sup>11,25–27</sup> Most of the host–guest attractions transiently formed are noncovalent and weak, so that the adsorbent potential in a subnanopore is shallow and anharmonic. These inherent complexities make nanoporous host–guest science challenging and fascinating.

Salts of the  $\text{Li}^+$  endohedral buckminsterfullerene ( $\text{Li}^+\text{@C}_{60}$ ), which crystallize along with counteranions, have recently been recognized as host–guest systems ideal for studying the nuclear translational motion of  $\text{Li}^+$  under subnanoscale confinement.<sup>28–34</sup> Although a  $\text{Li}^+\text{@C}_{60}$  salt does not possess any continuous transport pathways for  $\text{Li}^+$ , there remain translational degrees of  $\text{Li}^+$  freedom within every  $\text{C}_{60}$  cage. In an isolated cluster model of  $\text{Li}^+\text{@C}_{60}$ , or  $\text{Li}^+\text{@C}_{60}^-$ , the icosahedral ( $I_h$ ) cage and the  $\pi$  electron cloud create a polarizable and anharmonic spherical-hollow potential for  $\text{Li}^+$  and potential wells therein, specifically underneath the centers of the individual  $\text{C}_6$  and  $\text{C}_5$  rings.<sup>35–38</sup> Zhang *et al.* theoretically revealed that although  $\text{Li}^+$  encounters an energy barrier when moving between any two adjacent  $\text{C}_6$  potential wells, quantum tunneling brings about superpositions of the degenerate

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**Fig. 1** Structure of the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (6 K) model, with all the structural parameters determined by an X-ray diffraction experiment at 6 K.<sup>31</sup> (a) side view and (b) top view of the north pole. A  $\text{Li}^+$  ion moves within the  $C_{60}$  cage. Underneath the  $C_6(1)$  ring and the  $C_6(1')$  ring, indicated by the yellow highlights, there is an orange point and a maroon point, respectively. These points represent the two disordered sites of  $\text{Li}^+$  at 6 K,<sup>31</sup> both of which are on the  $C_3$  axis aligned with the  $z$  axis. The inversion-symmetry-breaking distortion below  $T_C$  renders the northern ( $z > 0$ ) and southern ( $z < 0$ ) hemispheres of the cage asymmetrical; here we note that the origin of the Cartesian  $xyz$  frame is placed at the  $C_{60}$  cage center but shifted in the panels for clarity. As a result, the two disordered sites are symmetrically nonequivalent and have different occupancies of  $\text{Li}^+$ ; that is, the northern disordered site underneath the  $C_6(1)$  ring (*i.e.*, the orange point) has a high occupancy of 0.76, whereas the southern one underneath the  $C_6(1')$  ring

basis states of  $\text{Li}^+$  centered on each  $C_6$  well.<sup>38</sup> As a result, the nuclear ground state of  $\text{Li}^+$  is a symmetric (in-phase) superposition state evenly localized over each of the 20  $C_6$  wells. When the cage structure is slightly distorted from the  $I_h$  symmetry, the resultant perturbation of the potential wells can dramatically localize the ground state in a single  $C_6$  well.<sup>38</sup> A question then arises as to whether such cage-distortion effects on the adsorbent potential and the  $\text{Li}^+$  motion persist in a real  $\text{Li}^+@C_{60}$  salt, not only in the cluster models.

Aoyagi *et al.* obtained a variety of  $\text{Li}^+@C_{60}$  salts, which have crucially different distributions of  $\text{Li}^+$ .<sup>28–31</sup> Among the studied salts,  $[\text{Li}^+@C_{60}]\text{PF}_6^-$  is believed to most evidently manifest the nuclear quantum nature of  $\text{Li}^+$ .<sup>30–34</sup> The salt has a rock-salt-type cubic structure (Fig. S1, ESI†), in which every  $\text{Li}^+@C_{60}$  cluster is almost octahedrally coordinated by six  $\text{PF}_6^-$  anions.<sup>30,31</sup> In the temperature range of  $24\text{ K} < T < 100\text{ K}$ , each  $C_{60}$  cage has  $S_6$  symmetry, not the ideal  $I_h$  symmetry, due to weak cage elongation along the  $S_6$  axis (see, for example, Fig. 1 in our previous paper<sup>25</sup> for the local structure at 40 K). The two  $C_6$  rings located parallel to and diametrically opposite one another along the axis (*i.e.*, the rings designated as  $C_6(1)$  in the paper<sup>25</sup>) are both slightly pulled out from the inversion center  $i$  at the  $C_{60}$  cage center. A  $\text{Li}^+$  ion is localized at two equivalent (antipodal) disordered sites on the  $S_6$  axis, underneath each  $C_6(1)$  ring, with the same occupancy.<sup>30,31</sup> Aoyagi *et al.* ascribed this two-site disorder to quantum tunneling between the two  $C_6(1)$  potential wells deepened by the electrostatic interactions between  $\text{Li}^+$  and the neighboring  $\text{PF}_6^-$  anions.<sup>30,31</sup> This provides a new perspective on the control of endohedral confinement by intermolecular interactions.<sup>28–31</sup> On the other hand, we have recently attributed the two-site disorder primarily to the  $C_{60}$  cage elongation,<sup>25</sup> which is relatively consistent with the aforementioned confinement control by cage distortion.<sup>38</sup> We demonstrated that the  $\text{Li}^+ \cdots \text{PF}_6^-$  electrostatic attraction has only a subdominant effect on the energy landscape of the adsorbent potential. Even when disregarding the attraction, the  $C_{60}$  cage elongation itself sterically and electrostatically makes the  $C_6(1)$  wells deepest.<sup>25</sup> Consequently, the calculated nuclear ground state of  $\text{Li}^+$  is nearly twofold degenerate and evenly localized over the two  $C_6(1)$  wells near the experimental disordered sites at 40 K.<sup>25</sup> Our findings suggest that without a thorough analysis, one cannot identify the factors determining the endohedral motion of  $\text{Li}^+$  in a  $\text{Li}^+@C_{60}$  salt.

As the temperature decreases below  $T_C = 24\text{ K}$ ,  $[\text{Li}^+@C_{60}]\text{PF}_6^-$  exhibits an antiferroelectric phase transition involving the two-site disorder–order transition of every  $\text{Li}^+$  ion (Fig. S1, ESI†).<sup>31</sup> Fig. 1 shows the local structure of the X-ray crystallography data at 6 K.<sup>31</sup> The structure, for example, the  $C_{60}$  cage orientation and the  $\text{PF}_6^-$  coordination, is fairly similar to that at 40 K. However, the  $C_{60}$  cage loses the inversion symmetry due to weak distortion,<sup>31</sup> which lowers the  $S_6$

(*i.e.*, the maroon point) has a low occupancy of 0.24. The ring centers denoted with (without) a prime are in the southern (northern) hemisphere. The unwrapped geometry of the cage is presented in Fig. S2 (ESI†).

symmetry to a subgroup,  $C_3$ . The  $S_6$  axis of the 40 K structure is reduced to the  $C_3$  axis aligned with the polar  $z$  axis in Fig. 1, and there appears to be a subtle asymmetry between the northern ( $z > 0$ ) and southern ( $z < 0$ ) hemispheres of the cage. For example, the two  $C_6(1)$  centers in the 40 K structure become nonequivalent in the 6 K structure, namely the  $C_6(1)$  center at the north pole and the  $C_6(1')$  center at the south pole (see the two rings indicated by yellow highlights in Fig. 1). Accordingly, every  $\text{Li}^+$  ion becomes strongly localized at one of the two nonequivalent disordered sites on the  $C_3$  axis.<sup>31</sup> The northern disordered site underneath the  $C_6(1)$  ring has a high occupancy of 0.76, whereas the southern one underneath the  $C_6(1')$  ring has a low occupancy of 0.24 (the orange point and the maroon point, respectively, in Fig. 1). Aoyagi *et al.* attributed this two-site disorder–order transition to the antiferroelectric interactions of local electric dipole moments, which arise from every  $\text{Li}^+$  ion localized in coordination spheres comprised of  $\text{PF}_6^-$  anions.<sup>31</sup>

Suzuki *et al.* pointed out that several absorption peaks in the terahertz (THz) frequency range would serve as fingerprints of the endohedral motion of  $\text{Li}^+$  in  $[\text{Li}^+@C_{60}]\text{PF}_6^-$ .<sup>32,33</sup> At 300 K, there exists a broad peak at around 1.2 THz, which was assigned to the hindered rotations of  $\text{Li}^+$  in the cage. As the temperature decreases below 120 K, the broad peak gradually decreases in intensity, and then a new peak appears at 2.2 THz. They attributed the peak disappearance and appearance to low-temperature suppression of the hindered rotations and accompanying localization at the two disordered sites, respectively.<sup>32</sup> Below 30 K, a weak peak appears at 2.65 THz, which was ascribed to the two-site disorder–order transition at  $T_C$ .<sup>32</sup> The overall mechanism is consistent with that proposed by Aoyagi *et al.*<sup>31</sup> However, low-lying nuclear energy levels and the relevant motions of  $\text{Li}^+$ , which reflect the specific endohedral environment<sup>25,38</sup> and underlie the low-temperature THz spectra, deserve further investigation. For instance, before we revealed that the energy-level structure above  $T_C$  essentially depends on the  $C_{60}$  cage elongation, symmetry, and multiple-well features of the potential,<sup>25</sup> it had been assumed to be governed by the rigid-rotor angular momenta with *a posteriori* corrections.<sup>32</sup> It has yet to be clarified how the transition at  $T_C$  changes the energy-level structure.

Focusing on the local structural distortion and intermolecular interactions in  $[\text{Li}^+@C_{60}]\text{PF}_6^-$  at 6 K, we aim to discuss how they are associated with the endohedral motion of  $\text{Li}^+$ , particularly the two-site disorder–order transition, and theoretically identify the spectroscopic fingerprints of the transition. With the aid of the resolution-of-the-identity second-order Møller–Plesset perturbation (RI-MP2) theory<sup>39</sup> and modified Morse pair potentials,<sup>40</sup> we first developed an accurate potential energy function for the  $\text{Li}^+$  motion within a local structure model of the salt. We then obtained hundreds of low-energy nuclear wave functions of  $\text{Li}^+$  using the Fourier grid Hamiltonian (FGH) method.<sup>41–44</sup> Our local structure model does not incorporate the crystal periodicity. Thus, the model incorporates neither the antiferroelectric interactions nor the collective ordering of the  $\text{Li}^+$  ions in the salt. Nevertheless,

the wave function of the nuclear ground state, as well as those of several low-energy excited states, has a large amplitude over the deep  $C_6(1)$  well near the high-occupancy disordered site at 6 K. This demonstrates that the weak inversion-symmetry-breaking distortion of the local structure affects the intermolecular interactions, thereby indeed correlating with the two-site disorder–order transition. The experimental THz spectra below and above  $T_C$  are qualitatively reproduced by our theoretical results, in which the disorder–order transition is evident as blueshifts of two absorption peaks associated with the localized ground state.

## II. Theoretical models and methods

We study the  $\text{Li}^+$  behavior in  $[\text{Li}^+@C_{60}]\text{PF}_6^-$ , particularly below  $T_C = 24$  K, in detail. In what follows, we outline how we obtained a local structure model of the salt, the potential energy function, and low-energy nuclear wave functions of  $\text{Li}^+$  confined within the model. The local structure models newly and previously<sup>25</sup> constructed are designated as  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (6 K) and  $[\text{Li}^+@C_{60}]40\text{PF}_6^-$  (40 K), respectively, since their geometries were determined using the X-ray crystal structures at 6 K ( $< T_C$ ) and 40 K ( $> T_C$ ).<sup>31</sup> Unlike the real salt, each model includes only six  $\text{PF}_6^-$  anions, which is indicated by the number 6 in the model name. The current methodology is an extension of our previous one,<sup>25</sup> designed to study the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (40 K) model with  $S_6$  symmetry, to the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (6 K) model with the lower  $C_3$  symmetry.

### A. The $[\text{Li}^+@C_{60}]6\text{PF}_6^-$ (6 K) model and the radial grid therein

Given the importance of the local structure and interactions in the endohedral  $\text{Li}^+$  motion,<sup>25</sup> we intend to accurately model them. We determined the geometry of the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (6 K) model (Fig. 1) by extracting a single  $\text{Li}^+@C_{60}$  cluster and six  $\text{PF}_6^-$  anions, which almost octahedrally coordinate the cluster, from the X-ray crystal structure at 6 K (Fig. S1, ESI†).<sup>31</sup> The model incorporates both the  $C_{60}$  cage distortion and the local intermolecular interactions in the real salt (*e.g.*, the  $\text{Li}^+ \cdots \text{PF}_6^-$  electrostatic attraction). When we disregard the unfixed position of  $\text{Li}^+$ , the whole model, the pristine  $C_{60}$  cage, and the cluster of the six  $\text{PF}_6^-$  anions all belong to the  $C_3$  point group. In contrast, they belong to the  $S_6$  point group at 40 K,<sup>25</sup> which indicates that the pseudo-octahedral  $\text{PF}_6^-$  coordination sphere, as well as the  $C_{60}$  cage, exhibits inversion-symmetry breaking at 6 K. We assumed that  $\text{Li}^+$  moves within the fixed model structure, considering the experimental observation that the rotations of the  $C_{60}$  cages and the  $\text{PF}_6^-$  anions are strongly restricted below 370 K.<sup>30</sup> Any interactions beyond the next nearest neighbors, including the proposed antiferroelectric interactions,<sup>31</sup> were ignored in the model. The  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (40 K) model had been constructed in a similar fashion.<sup>25</sup>

We defined the Cartesian  $xyz$  frame, shown in Fig. 1, as follows. In our previous paper,<sup>25</sup> the origin of the frame is placed at the center of mass of the  $C_{60}$  cage, or the inversion center  $i$ , of the  $[\text{Li}^+@C_{60}]40\text{PF}_6^-$  (40 K) model. The  $z$  axis is

aligned with the  $S_6$  axis, on which the two equivalent disordered sites of  $\text{Li}^+$  are located ( $z = \pm 1.348 \text{ \AA}$ ). The  $x$  axis points in the direction of the center of a next-nearest-neighbor  $\text{C}_{60}$  cage in the X-ray crystal structure at 40 K. Supposing that the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (40 K) model is clamped onto this  $xyz$  frame as a reference geometry, we arranged the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (6 K) model in the frame in such a way that the displacement vector of the  $\text{C}_{60}$  cage satisfies the center-of-mass and static Eckart conditions.<sup>45</sup> In other words, the translations and rotations of the  $\text{C}_{60}$  cage were almost eliminated from the displacement between the two geometries. This elimination will later enable us to address the question of how the  $\text{C}_{60}$  cage distortion with the inversion-symmetry breaking, or  $S_6 \rightarrow C_3$  symmetry lowering, affects the adsorbent potential and, thus, the nuclear wave functions of  $\text{Li}^+$  through superpositions. The displacement of each C atom is  $0.003 \text{ \AA}$  on average, indicating how weak the distortion is. The displacement of each atom of the  $\text{PF}_6^-$  ions is also small (*i.e.*,  $0.018 \text{ \AA}$  on average). The  $C_3$  axis of the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (6 K) model is aligned with the  $z$  axis, on which the disordered sites at 6 K are located (Fig. 1). The high-occupancy disordered site at the north pole (the orange point underneath the  $\text{C}_6(1)$  ring in Fig. 1) and the low-occupancy one at the south pole (the maroon point underneath the  $\text{C}_6(1')$  ring in Fig. 1a) are  $1.436$  and  $1.366 \text{ \AA}$  apart from the  $\text{C}_{60}$  cage center at the origin, respectively. The high-occupancy disordered site is somewhat pulled out from the cage center as compared to the disordered sites at 40 K. The spherical coordinates  $(r, \theta, \phi)$  were defined with respect to the  $xyz$  frame, where  $\theta \in [0^\circ, 180^\circ]$  and  $\phi \in [0^\circ, 360^\circ]$ .

The nuclear wave function of  $\text{Li}^+$  is governed by the following Hamiltonian:

$$\hat{H} = -\frac{\hbar^2}{2m} \left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) + V(x, y, z), \quad (1)$$

where  $m$  is  $7.016$  amu, the mass of  $^7\text{Li}$ , and the coordinates  $(x, y, z)$  specify its position. Preparatory to determining the potential energy function  $V(x, y, z)$  for the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (6 K) model, we defined a radial grid using 62 one-dimensional radial rays emanating from the  $\text{C}_{60}$  cage center ( $r = 0 \text{ \AA}$ ). Different rays pass through different representative points on the cage surface of  $\text{C}_{60}$ . The 62 representative points defined are symmetrically nonequivalent to each other, and their  $C_3$  symmetry operations yield a total of 182 points distributed to cover the whole cage surface. The representative points consist of  $\text{C}_6$  ring centers,  $\text{C}_5$  ring centers, C atoms, centers of C–C bonds shared by a  $\text{C}_5$  ring and a  $\text{C}_6$  ring, and centers of C=C bonds shared by two  $\text{C}_6$  rings. The representative points in the northern hemisphere include four  $\text{C}_6$  centers ( $\text{C}_6(1), \dots, \text{C}_6(4)$ ), two  $\text{C}_5$  centers ( $\text{C}_5(1), \text{C}_5(2)$ ), and ten C atoms ( $\text{C}^1, \dots, \text{C}^{10}$ ).<sup>46</sup> The numbers in parentheses or superscript indicate the nonequivalent points (Fig. 1 and Fig. S2, ESI<sup>†</sup>); in other words, the same symbol is used for a representative point and its equivalent points. We indicate the rings, the rays, and the potential wells associated with a representative point using its symbol (*e.g.*, the  $\text{C}_6(1)$  ray). In the southern hemisphere also, there are representative points at four  $\text{C}_6$  centers ( $\text{C}_6(1'), \dots, \text{C}_6(4')$ ), two

$\text{C}_5$  centers ( $\text{C}_5(1'), \text{C}_5(2')$ ), and ten C atoms ( $\text{C}^{1'}, \dots, \text{C}^{10'}$ ).<sup>46</sup> Any points denoted with a prime are in the southern hemisphere. Two points denoted with the same symbol except for the prime (*e.g.*,  $\text{C}_6(1)$  and  $\text{C}_6(1')$ ) are equivalent in the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (40 K) model but become nonequivalent in the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (6 K) model due to the inversion-symmetry breaking. The symbols defined here are the same as those used for the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (40 K) model<sup>25</sup> if the prime symbol is ignored. The remaining representative points are 20 C–C centers and 10 C=C centers, which are specified with the symbols for the participating C atoms (*e.g.*, the  $\text{C}^1\text{--C}^2$  centers). We evenly spaced 32 grid points on every radial ray in addition to the grid point at the origin ( $r = 0 \text{ \AA}$ ). The step size  $\Delta r$  is  $0.0625 \text{ \AA}$ . We took  $\Delta r$  to be half of the previously employed value<sup>25</sup> to properly describe the subtle asymmetry between the northern and southern hemispheres. The radial grid thus obtained consists of 1985 grid points, corresponding to 5825 grid points when symmetry is considered.

We explain the geometrical configuration of the individual  $\text{C}_6$  and  $\text{C}_5$  rings in the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (6 K) model. In the northern hemisphere (Fig. 1 and Table S1, ESI<sup>†</sup>), a  $\text{C}_6(i)$  (or  $\text{C}_5(i)$ ) center with a smaller  $i$  is closer to the polar  $z$  axis and has an inclination angle nearer to  $0^\circ$  (*i.e.*, the north pole). Similarly, in the southern hemisphere (Fig. 1a and Table S2, ESI<sup>†</sup>), a  $\text{C}_6(i')$  (or  $\text{C}_5(i')$ ) center with a smaller  $i$  is closer to the polar  $z$  axis and has an inclination angle nearer to  $180^\circ$  (*i.e.*, the south pole), except for the  $\text{C}_6(3')$  and  $\text{C}_6(4')$  centers at almost the same angle.<sup>47</sup> In short, a small  $i$  indicates that the corresponding ring is a near-pole ring. The  $\text{C}_6(1)$  ring at the north pole is adjoined by three  $\text{C}_6(2)$  rings and three  $\text{C}_5(1)$  rings (Fig. 1b). Every  $\text{C}_6(2)$  ring is pointed toward by a single F atom of  $\text{PF}_6^-$  (Fig. 1b) and adjoined by a  $\text{C}_5(2)$  ring (Fig. 1a). The  $\text{C}_6(1')$ ,  $\text{C}_6(2')$ ,  $\text{C}_5(1')$ , and  $\text{C}_5(2')$  rings are arranged in a similar way in the southern hemisphere. The remaining  $\text{C}_6(3)$ ,  $\text{C}_6(4)$ ,  $\text{C}_6(3')$ , and  $\text{C}_6(4')$  rings are arranged near the equatorial plane, forming a zigzag band that winds around the polar  $z$  axis (Fig. 1a and Fig. S2, ESI<sup>†</sup>). The configuration of all  $\text{C}^j$  atoms ( $j = 1, \dots, 10, 1', \dots, 10'$ ) is shown in an unwrapped map of the  $\text{C}_{60}$  cage (Fig. S2, ESI<sup>†</sup>).

## B. The RI-MP2 potential energy surface for the endohedral motion of $\text{Li}^+$ and its modeling

The radial grid covers a large part of the cage interior of the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (6 K) model. To determine the potential energy function  $V(x, y, z)$  for the electronic ground state, we performed single-point RI-MP2 calculations<sup>39</sup> for 1985 geometries of the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (6 K) model, in each of which  $\text{Li}^+$  is fixed at a different grid point. We employed the Cartesian cc-pVDZ basis set<sup>48–50</sup> and the auxiliary basis set optimized for it<sup>51</sup> and included the  $1s$  core orbital of Li in the set of correlated valence orbitals to consider the dynamical correlation effects arising from the  $1s$  electrons. We used GAMESS-US (2019 R2) to perform these calculations.<sup>51</sup> Taking the RI-MP2 energy of the geometry with  $\text{Li}^+$  at the  $\text{C}_{60}$  cage center (origin) to be zero, we proceeded to fit a model function to the relative RI-MP2 energies,  $\Delta E_{\text{RI-MP2}}$ , on the grid. The lowest energy,  $\Delta E_{\text{RI-MP2}} = -10.73 \text{ kcal mol}^{-1}$ , was found at  $r = 1.4375 \text{ \AA}$  on the  $\text{C}_6(1)$  ray.<sup>52</sup>



Because we aim to examine the  $\text{Li}^+$  behavior at low temperatures throughout this study, we selected the 637 lowest-energy grid points, at any of which  $\Delta E_{\text{RI-MP2}}$  is below  $-7.0 \text{ kcal mol}^{-1}$ . To the grid data selected, we fitted the sum of the modified-Morse  $\text{Li}^+-\text{C}$  pair potentials,<sup>40</sup> in which the same  $\text{Li}^+-\text{C}^j$  pair potential ( $j = 1, \dots, 10, 1', \dots, 10'$ ) is employed for the three equivalent  $\text{C}^j$  atoms ( $k = 1, 2$ , and  $3$ ):

$$V_{\text{es+exrep+pol+disp}}^{6\text{K}}(x, y, z) = \sum_{j \in \{1, \dots, 10, 1', \dots, 10'\}} \sum_{k=1}^3 \left\{ D_e^j \left( \frac{\exp[2\alpha_e^j s_e^j (1 - (s_k^j/s_e^j)^\zeta)]/\zeta - 2 \exp[\alpha_e^j s_e^j (1 - (s_k^j/s_e^j)^\zeta)]/\zeta - A^j}{1 + A^j} \right) + V_{\text{shift}} \right\}, \quad (2)$$

with  $s_k^j$  being the interatomic distance between  $\text{Li}^+$  and a  $\text{C}^j$  atom of the  $\text{C}_{60}$  cage. While  $D_e^j$ ,  $\alpha_e^j$ , and  $s_e^j$  are the Morse parameters, the additional  $\zeta$  makes the long- and short-range profiles of the pair potential flexible as compared to the conventional Morse potential.<sup>40</sup> The baseline shift  $V_{\text{shift}}$ , as well as  $A^j$  defined by

$$A^j = \begin{cases} 0, & \text{if } \zeta \geq 0 \\ \exp[2\alpha_e^j s_e^j/\zeta] - 2 \exp[\alpha_e^j s_e^j/\zeta], & \text{otherwise,} \end{cases} \quad (3)$$

enables the potential energy function in eqn (2) to approach zero at the  $\text{C}_{60}$  cage center, as with  $\Delta E_{\text{RI-MP2}}$ . The potential energy function is denoted by  $V_{\text{es+exrep+pol+disp}}^{6\text{K}}(x, y, z)$ , instead of  $V(x, y, z)$ , to remind the reader that it effectively includes all four types of intermolecular interactions in the  $[\text{Li}^+@ \text{C}_{60}]6\text{PF}_6^-$  (6 K) model, namely the electrostatic (es), exchange repulsion (exrep), polarization (pol), and dispersion (disp) interactions in a standard energy decomposition analysis (EDA) scheme.<sup>53</sup> For instance,  $V_{\text{es+exrep+pol+disp}}^{6\text{K}}(x, y, z)$  effectively includes the interactions caused by the pseudo-octahedral  $\text{PF}_6^-$  coordination (e.g., the  $\text{Li}^+ \cdots \text{PF}_6^-$  electrostatic attraction)<sup>25</sup> through the fit to the  $\Delta E_{\text{RI-MP2}}$  grid data, although the functional form itself is comprised only of the  $\text{Li}^+-\text{C}$  pair potentials. Besides,  $V_{\text{es+exrep+pol+disp}}^{6\text{K}}(x, y, z)$  and the Hamiltonian in eqn (1), with  $V(x, y, z)$  replaced by  $V_{\text{es+exrep+pol+disp}}^{6\text{K}}(x, y, z)$ , properly possess the  $\text{C}_3$  symmetry of the system.

It turns out that  $V_{\text{es+exrep+pol+disp}}^{6\text{K}}(x, y, z)$  fits the  $\Delta E_{\text{RI-MP2}}$  grid data reasonably well; see Table S3 (ESI†) for the parameters optimized by the Levenberg–Marquardt (LM) least-squares fitting.<sup>54,55</sup> The root-mean-square error (RMSE) is  $0.057 \text{ kcal mol}^{-1}$ . The RMSE value remains small when recalculating it by including  $\Delta E_{\text{RI-MP2}}$  grid data beyond  $-7.0 \text{ kcal mol}^{-1}$  (Table S4, ESI†). The potential energy at the  $\text{C}_{60}$  cage center,  $-0.772 \text{ kcal mol}^{-1}$ , is close to zero. We hence conclude that  $V_{\text{es+exrep+pol+disp}}^{6\text{K}}(x, y, z)$  is reliable not only in the low energy region considered in the LM fitting procedure but also in a somewhat higher energy region.

### C. Energy decomposition analysis

To examine the intermolecular interactions, we divided the entire  $[\text{Li}^+@ \text{C}_{60}]6\text{PF}_6^-$  (6 K) model into three closed-shell fragments, namely  $\text{Li}^+$ ,  $\text{C}_{60}$ , and the cluster of the six  $\text{PF}_6^-$  anions,

and imposed the assumption of frozen intrafragment geometries. Using the Kitaura–Morokuma program<sup>51,56,57</sup> and following the philosophy of the EDA scheme proposed by Su *et al.*,<sup>53</sup> we decomposed the RI-MP2 interaction energy of the three fragments into the electrostatic interaction energy, the exchange repulsion energy, the polarization interaction energy, and the dispersion interaction energy as follows:

$$E_{\text{RI-MP2}}(x, y, z) = \sum_{i=1}^3 E_{\text{RI-MP2}}^{\text{ith fragment}} \\ = E_{\text{es}}(x, y, z) + E_{\text{exrep}}(x, y, z) + E_{\text{pol}}(x, y, z) + E_{\text{disp}}(x, y, z), \quad (4)$$

where  $E_{\text{RI-MP2}}(x, y, z)$  is the total energy of the  $[\text{Li}^+@ \text{C}_{60}]6\text{PF}_6^-$  (6 K) model with  $\text{Li}^+$  fixed at  $(x, y, z)$  and  $E_{\text{RI-MP2}}^{\text{ith fragment}}$  is the energy of the  $i$ th fragment. The electrostatic term,  $E_{\text{es}}$ , and the exchange repulsion term,  $E_{\text{exrep}}$ , are associated with quasi-classical Coulomb interactions among the fragments' charge distributions and the Pauli exclusion, respectively.<sup>56,57</sup> The polarization term,  $E_{\text{pol}}$ , is an orbital relaxation term that incorporates not only the intrafragment polarization but also the interfragment charge transfer.<sup>53</sup> The dispersion term,  $E_{\text{disp}}$ , arises from the dynamical electron correlation.<sup>53</sup> Taking the geometry with  $\text{Li}^+$  at the  $\text{C}_{60}$  cage center (origin) to be the reference, one obtains

$$\Delta E_{\text{RI-MP2}}(x, y, z) = \Delta E_{\text{es}}(x, y, z) + \Delta E_{\text{exrep}}(x, y, z) + \Delta E_{\text{pol}}(x, y, z) \\ + \Delta E_{\text{disp}}(x, y, z), \quad (5)$$

where  $\Delta$  denotes that every energy corresponds to an energy gain/loss due to the off-centering of  $\text{Li}^+$ . See our previous paper for further details.<sup>25</sup>

### D. Nuclear wave functions of the endohedral $\text{Li}^+$

Using the FGH method,<sup>41–44</sup> we numerically solved the time-independent Schrödinger equation with the Hamiltonian in eqn (1), with  $V(x, y, z)$  replaced by  $V_{\text{es+exrep+pol+disp}}^{6\text{K}}(x, y, z)$ , to obtain low-energy nuclear wave functions of  $\text{Li}^+$  in the  $[\text{Li}^+@ \text{C}_{60}]6\text{PF}_6^-$  (6 K) model. We employed the FGH basis set associated with a three-dimensional, uniform, Cartesian grid with  $101^3$  grid points. In the range of  $-3.45$  to  $3.45 \text{ Å}$  in each of the  $x$ ,  $y$ , and  $z$  directions, we evenly spaced 101 points and defined 101 pseudo-spectral states accordingly. The FGH basis set consists of the direct product states of these states, thus spatially localized at each of the Cartesian grid points and spanning a Hilbert space of dimension  $101^3$ . Before diagonalizing the Hamiltonian matrix, all of the potential energy elements beyond  $1.0 \times 10^3 \text{ kcal mol}^{-1}$  were linearly scaled to the range of  $1.0 \times 10^3$  to  $2.0 \times 10^3 \text{ kcal mol}^{-1}$ . This artificial scaling enabled us to avoid floating point overflows without any

marked effects on the low-energy eigensolutions of interest.<sup>25</sup> We diagonalized the Hamiltonian matrix using the thick-restart Lanczos (TRLan) method,<sup>58,59</sup> yielding the 802 lowest-energy nuclear wave functions of  $\text{Li}^+$ . The TRLan method, unlike the standard Lanczos method, can be safely applied to degenerate eigenproblems<sup>59</sup> and is hence suitable for our study of a high-symmetry porous system.

We also employed the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (40 K) model and obtained the 801 lowest-energy nuclear wave functions of  $\text{Li}^+$ . The theoretical methods and computational details of the FGH/TRLan calculation, as well as the potential energy function  $V_{\text{es+exrep+pol+disp}}^{40\text{K}}(x,y,z)$ ,<sup>60</sup> were the same as those previously reported.<sup>25</sup> This calculation, essentially a recalculation of our previous results,<sup>61</sup> was required to obtain THz absorption spectra above  $T_C$ . The THz spectrum calculation was performed in accordance with the approach by Zhang *et al.*<sup>38</sup> Its details will be given later.

### III. Results and discussion

#### A. Inversion-symmetry breaking of the $C_{60}$ cage

For the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (40 K) model, we previously examined the distance of each C atom from the  $C_{60}$  cage center.<sup>25</sup> The distance,  $r_C$ , is a good measure of the  $C_{60}$  cage distortion from  $I_h$  symmetry. The  $r_C$  distances are shown as a bubble chart in Fig. S3a (ESI†). On the whole, C atoms near the north and south poles have long  $r_C$  distances, indicating the  $C_{60}$  cage elongation along the  $S_6$  axis. All members of the two  $C_6(1)$  rings at the poles are pulled out from the cage center ( $r_C = 3.545$  Å), whereas four members of every  $C_5(2)$  ring near the equatorial  $xy$  plane are pushed toward the cage center ( $r_C < 3.539$  Å, Fig. S3a in ESI†).<sup>25</sup> The remaining member of each  $C_5(2)$  ring, the  $C^9$  atom, is an exception, having a long  $r_C$  distance of 3.544 Å.<sup>25</sup> The mean distance of  $r_C$  is 3.540 Å, with a standard deviation of 0.0045 Å. The difference between the longest and shortest  $r_C$  distances is 0.014 Å. Although this  $I_h \rightarrow S_6$  distortion of  $C_{60}$  is weak, it strongly affects the depths of potential wells and causes the localization of low-energy nuclear wave functions of  $\text{Li}^+$ .<sup>25</sup>

With the  $C_{60}$  cage center fixed at the origin, we analyzed the inversion-symmetry-breaking  $S_6 \rightarrow C_3$  distortion of the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (6 K) model in terms of the  $r_C$  distances. For the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (6 K) model, we separately list the  $r_C$  distances for the northern hemisphere (Table S5, ESI†) and those for the southern hemisphere (Table S6, ESI†). The bubble chart of the  $r_C$  distances is given in Fig. S3b (ESI†). Please remember that the north pole is at the bottom ( $\theta = 0^\circ$ ) and the south pole is at the top ( $\theta = 180^\circ$ ) in the bubble chart and in all polar plots in this paper. Among all C atoms, the  $C^1$  and  $C^2$  atoms, the members of the  $C_6(1)$  ring at the north pole, are the first and second farthest from the cage center ( $r_C = 3.550$  and 3.549 Å, respectively). The  $C^{1'}$  and  $C^{2'}$  atoms, the members of the  $C_6(1')$  ring at the south pole, are somewhat far from the cage center ( $r_C = 3.543$  Å), but not as far as the  $C^1$  and  $C^2$  atoms are. All the members of the  $C_5(2)$  and  $C_5(2')$  rings near the equator, except for the  $C^9$  atoms, are close to the cage center

( $r_C < 3.541$  Å, Tables S5 and S6 in ESI†). The mean distance of  $r_C$  remains nearly unchanged (3.540 and 3.539 Å in the northern and the southern hemisphere, respectively), compared with that of the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (40 K) model (3.540 Å). Meanwhile, the standard deviation of  $r_C$  is larger in the northern hemisphere (0.0069 Å) and smaller in the southern hemisphere (0.0038 Å) than in the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (40 K) model. The difference between the longest and shortest  $r_C$  distances is also larger in the northern hemisphere (0.021 Å) and smaller in the southern hemisphere (0.012 Å) than in the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (40 K) model. Hence, while keeping the mean distance of  $r_C$  unchanged, the  $S_6 \rightarrow C_3$  symmetry lowering increases (reduces) the amount of variation in the  $r_C$  distances, that is, the degree of the  $C_{60}$  cage distortion from  $I_h$  symmetry, in the northern (southern) hemisphere. This feature can be qualitatively explained using the bubble chart of  $\Delta r_C = r_C^{6\text{K}} - r_C^{40\text{K}}$  (Fig. S4, ESI†), the  $r_C$  change caused by the  $S_6 \rightarrow C_3$  symmetry lowering. The chart shows an inversion-asymmetric pattern with four horizontal stripes, alternating between blue and red. The pattern indicates that  $\Delta r_C$  is positive, negative, slightly positive, and negative for C atoms in the region near the north pole, north of the equator, south of the equator, and near the south pole, respectively. Therefore, the  $S_6 \rightarrow C_3$  symmetry lowering increases (reduces) the differences between the long  $r_C$  distances of near-pole C atoms and the short  $r_C$  distances of near-equator C atoms in the northern (southern) hemisphere. The  $S_6 \rightarrow C_3$  distortion causes the  $C_6(1)$  ring at the north pole (the  $C_6(1')$  ring at the south pole) to be further pulled out from (conversely, pushed back toward) the cage center (Fig. S4, ESI†). The  $C_6(1)$  ring thus becomes the farthest from the cage center among all  $C_6$  rings. Let us note that all of the analysis in Section IIIA pertains to the experimental structures<sup>31</sup> as well as our model structures.

#### B. Potential energy function and potential wells for the endohedral $\text{Li}^+$

The potential energy function of the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (40 K) model,  $V_{\text{es+exrep+pol+disp}}^{40\text{K}}$ , has the shape of an approximately spherical hollow shell with a radius of  $r \approx 1.375$  Å near the  $C_{60}$  cage.<sup>25</sup> The potential wells are on the shell, specifically underneath some of the ring centers. The relative stabilities of the energy minima with respect to  $r$  along the ring's center rays are summarized as follows (in increasing energetic order):  $C_6(1) < C_6(2) < C_6(3) \sim C_6(4) < C_5(1) < C_5(2)$ .<sup>25</sup> This suggests that there are deep potential wells underneath the  $C_6$  centers rather than the  $C_5$  ones. In addition, a deeper well is located underneath a  $C_6$  (or  $C_5$ ) center that is closer to a pole. The two  $C_6(1)$  wells at the poles, near the disordered sites at 40 K, are the deepest chiefly because the  $C_6(1)$  rings are noticeably pulled out from the cage center.<sup>25</sup> The  $C_5(1)$  wells are only slightly less stable than the  $C_6(1)$  wells (by just 0.27 kcal mol<sup>-1</sup>) but are very shallow. Underneath the  $C_5(2)$  rings pushed toward the cage center, there are no wells that restrict the librational motion of  $\text{Li}^+$ .

As regards the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (6 K) model, Fig. 2 shows the potential energy function  $V_{\text{es+exrep+pol+disp}}^{6\text{K}}$  evaluated along the

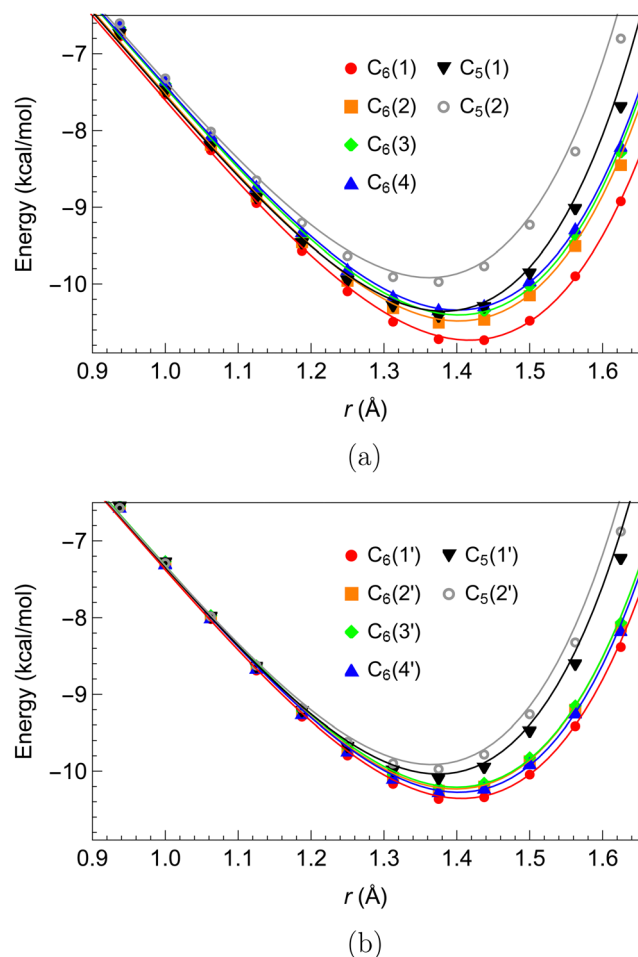


Fig. 2 Radial profiles of the potential energy function  $V_{\text{es+exrep+pol+disp}}^{\text{6K}}$  (curves) along the  $C_6$  and  $C_5$  rays in (a) the northern hemisphere ( $z > 0$ ) and (b) the southern hemisphere ( $z < 0$ ) of the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (6 K) model. The symbols are the single-point RI-MP2 energies,  $\Delta E_{\text{RI-MP2}}$ , which are relative to the energy at  $r = 0$  Å (i.e., the  $C_{60}$  cage center). See Fig. S5 and S6 (ESI†) for the other radial rays.

$C_6$  and  $C_5$  rays (curves), together with the corresponding  $\Delta E_{\text{RI-MP2}}$  grid data (symbols). In both the northern hemisphere (Fig. 2a) and the southern hemisphere (Fig. 2b), the radial energy profiles are well reproduced by the model function. All the rays, including non-ring-center rays, exhibit energy minima around  $r = 1.375$  Å (Fig. 2; Fig. S5, S6 and Table S7, ESI†). Thus, we show the polar (angular) plots of  $V_{\text{es+exrep+pol+disp}}^{\text{6K}}$  and  $\Delta E_{\text{RI-MP2}}$  at  $r = 1.375$  Å in Fig. 3a and Fig. S7 (ESI†). The polar plots are similar to each other and suggest that there are deep potential wells underneath the  $C_6$  centers but no wells underneath the  $C_5(2)$  and  $C_5(2')$  ones. Compared to the near-equator sites underneath the  $C_5(2)$  and  $C_5(2')$  centers, the near-pole sites underneath the  $C_5(1)$  and  $C_5(1')$  ones are stable, but the wells there are shallow and neighbored by more stable  $C_6$  wells (see the rows labeled  $V_{\text{es+exrep+pol+disp}}^{\text{6K}}$  and  $\Delta E_{\text{RI-MP2}}$  in Table 1 for the relevant energies). The isosurface plots of  $V_{\text{es+exrep+pol+disp}}^{\text{6K}}$  (Fig. S8, ESI†) show an isolated island underneath each  $C_6$  center, a tied island underneath each of the  $C_5(1)$  and  $C_5(1')$  ones, and no islands underneath the  $C_5(2)$  and  $C_5(2')$  ones, which confirms the above

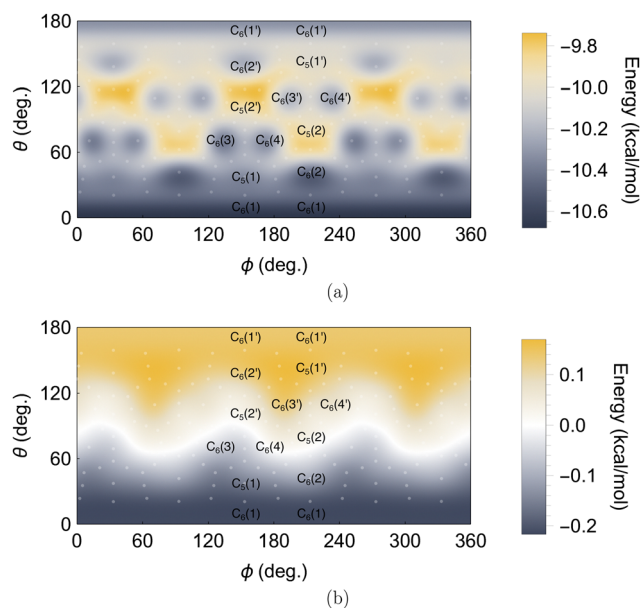


Fig. 3 Polar profiles of (a) the potential energy function  $V_{\text{es+exrep+pol+disp}}^{\text{6K}}$  for the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (6 K) model and (b) the energy change due to the  $S_6 \rightarrow C_3$  distortion (i.e.,  $V_{\text{es+exrep+pol+disp}}^{\text{6K}} - V_{\text{es+exrep+pol+disp}}^{\text{40K}}$ ). Both image maps are polar plots in a spherical shell with  $r = 1.375$  Å. The north pole and the south pole are spread out over all azimuth angles  $\phi$  at the bottom ( $\theta = 0^\circ$ ) and the top ( $\theta = 180^\circ$ ) of each map, respectively. The positions under the C atoms, bond centers, and ring centers of the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (6 K) model, except those under the  $C_6(1)$  and  $C_6(1')$  centers at the poles, are indicated by semitransparent white points. Given that the system has  $C_3$  symmetry, a minimum number of ring centers are indicated by their symbols. See Fig. S7 and S9 (ESI†) for the plots of the corresponding single-point calculation data.

observations. Hence, the key features of the adsorbent potential, specifically the spherical hollow structure of  $r \approx 1.375$  Å and the deep  $C_6$  wells therein, remain unchanged regardless of the distortion.

Yet, a closer look at the radial and polar plots reveals that the inversion-symmetry breaking affects the adsorbent potential. The radial energy profiles in the northern hemisphere (Fig. 2a and Fig. S5, ESI†) are notably different from those in the southern hemisphere (Fig. 2b and Fig. S6, ESI†), and the upper (southern) half and the lower (northern) half of the polar image are mutually inversion-asymmetric (Fig. 3a). The ring center wells in the northern (southern) hemisphere greatly (only slightly) differ in depth (Fig. 2 and 3a). Similar observations on the whole apply to the minima of the other rays in the two hemispheres (Fig. S5 and S6, ESI†); indeed, for any type of minima, the standard deviation of the depths is greater in the northern hemisphere than in the southern one (Table S7, ESI†). Besides, the inversion-symmetry breaking deepens the ring center wells in the northern hemisphere (i.e., the  $C_6(1), \dots, C_6(4)$  and  $C_5(1)$  wells) compared to the corresponding wells in the southern hemisphere (Table 1). In either of the rows labeled  $V_{\text{es+exrep+pol+disp}}^{\text{6K}}$  and  $\Delta E_{\text{RI-MP2}}$  in Table 1, the  $C_6(1)$ ,  $C_6(2)$ ,  $C_6(3)$ , and  $C_5(1)$  wells in the northern hemisphere are deeper than any wells in the southern hemisphere,<sup>62</sup> including the  $C_6(1')$  well at the south pole. The relative energy

**Table 1** Potential energies,  $V_{\text{es+exrep+pol+disp}}^{6\text{K}}$  and  $\Delta E_{\text{RI-MP2}}$ , and four interaction energies of the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (6 K) model. In all of the geometries examined,  $\text{Li}^+$  is fixed at  $r = 1.375$  Å underneath a ring center.<sup>a</sup> All energies are relative to the reference energies at  $r = 0$  Å and given in kcal mol<sup>-1</sup>. The mean distance between  $\text{Li}^+$  and the nearby ring's C atoms ( $d_{\text{Li-C}}$ ) is given in Å. See Tables S8 and S9 (ESI) for further decomposition into fragment-pair interaction energies

|                                            | Northern hemisphere ( $z > 0$ ) |          |          |          |          |            |
|--------------------------------------------|---------------------------------|----------|----------|----------|----------|------------|
|                                            | $C_6(1)$                        | $C_6(2)$ | $C_6(3)$ | $C_6(4)$ | $C_5(1)$ | $C_5(2)^b$ |
| $d_{\text{Li-C}}$                          | 2.355 (2.306) <sup>a</sup>      | 2.347    | 2.345    | 2.345    | 2.306    | 2.296      |
| $V_{\text{es+exrep+pol+disp}}^{6\text{K}}$ | -10.68 (-10.72)                 | -10.46   | -10.38   | -10.32   | -10.35   | -9.91      |
| $\Delta E_{\text{RI-MP2}}$                 | -10.72 (-10.73)                 | -10.51   | -10.40   | -10.34   | -10.43   | -9.97      |
| $\Delta E_{\text{es}}$                     | -0.87 (-0.90)                   | -0.89    | -0.45    | -0.39    | -0.09    | 0.32       |
| $\Delta E_{\text{exrep}}$                  | 11.50 (13.92)                   | 11.80    | 11.86    | 11.83    | 11.93    | 12.24      |
| $\Delta E_{\text{pol}}$                    | -20.76 (-23.12)                 | -20.78   | -21.01   | -20.98   | -21.30   | -21.39     |
| $\Delta E_{\text{disp}}$                   | -0.59 (-0.63)                   | -0.64    | -0.81    | -0.80    | -0.97    | -1.14      |

|                                            | Southern hemisphere ( $z < 0$ ) |           |           |           |           |             |
|--------------------------------------------|---------------------------------|-----------|-----------|-----------|-----------|-------------|
|                                            | $C_6(1')$                       | $C_6(2')$ | $C_6(3')$ | $C_6(4')$ | $C_5(1')$ | $C_5(2')^b$ |
| $d_{\text{Li-C}}$                          | 2.349                           | 2.345     | 2.345     | 2.346     | 2.303     | 2.298       |
| $V_{\text{es+exrep+pol+disp}}^{6\text{K}}$ | -10.33                          | -10.22    | -10.19    | -10.25    | -10.04    | -9.91       |
| $\Delta E_{\text{RI-MP2}}$                 | -10.36                          | -10.26    | -10.21    | -10.27    | -10.12    | -9.97       |
| $\Delta E_{\text{es}}$                     | -0.35                           | -0.44     | -0.11     | -0.20     | 0.42      | 0.36        |
| $\Delta E_{\text{exrep}}$                  | 11.79                           | 11.88     | 11.80     | 11.80     | 12.07     | 12.19       |
| $\Delta E_{\text{pol}}$                    | -21.18                          | -21.01    | -21.06    | -21.03    | -21.56    | -21.39      |
| $\Delta E_{\text{disp}}$                   | -0.63                           | -0.69     | -0.85     | -0.84     | -1.04     | -1.14       |

<sup>a</sup> For the  $C_6(1)$  well, we also investigated the geometry with  $\text{Li}^+$  at  $r = 1.4375$  Å. The results are presented in parentheses. <sup>b</sup> There are no potential wells underneath the  $C_5(2)$  and  $C_5(2')$  centers.

stabilization (destabilization) in the northern (southern) hemisphere occurs not only right underneath the ring centers but underneath non-ring-center points. This is clearly evident in the polar plot of  $V_{\text{es+exrep+pol+disp}}^{6\text{K}} - V_{\text{es+exrep+pol+disp}}^{40\text{K}}$  at  $r = 1.375$  Å (Fig. 3b), with the inversion-symmetric  $V_{\text{es+exrep+pol+disp}}^{40\text{K}}$  being the reference potential. Here,  $V_{\text{es+exrep+pol+disp}}^{6\text{K}} - V_{\text{es+exrep+pol+disp}}^{40\text{K}}$  is an inversion-asymmetric perturbative potential, which determines how the  $S_6 \rightarrow C_3$  distortion deforms the nuclear wave functions of  $\text{Li}^+$  through superpositions.<sup>63</sup> The polar plot in Fig. 3b shows a pattern composed of a northern gray stripe and a southern yellow stripe, which indicate the relative stabilization and destabilization, respectively. The degree of (de)stabilization is large near the poles. Similar features are present in the single-point RI-MP2 data (Fig. S9, ESI<sup>†</sup>). Underneath the  $C_6(1)$  ring, whose members have a positive  $\Delta r_{\text{C}}$  (*i.e.*, out-of-center, or outward, displacement),  $V_{\text{es+exrep+pol+disp}}^{6\text{K}} - V_{\text{es+exrep+pol+disp}}^{40\text{K}}$  is negative (*i.e.*, the well deepens).<sup>63</sup> In contrast, underneath the  $C_6(1')$  ring, whose members have a negative  $\Delta r_{\text{C}}$  (*i.e.*, inward displacement),  $V_{\text{es+exrep+pol+disp}}^{6\text{K}} - V_{\text{es+exrep+pol+disp}}^{40\text{K}}$  is positive (*i.e.*, the well becomes shallow).<sup>63</sup> These relationships are consistent with our previous finding that a deep well exists under a  $C_6$  ring pulled out from the  $C_{60}$  cage center.<sup>25</sup> In Fig. 3b, however, there is no four-stripe pattern in the bubble chart of  $\Delta r_{\text{C}}$  (Fig. S4, ESI<sup>†</sup>), which suggests that the inversion-asymmetric well deepening and shallowing cannot be solely attributed to the  $r_{\text{C}}$  changes. Similarly, the greater (lesser) amount of variation in the well depths in the northern (southern) hemisphere should not be entirely attributed to the strong (weak)  $C_{60}$  distortion from  $I_h$  symmetry, which we have discussed in Section IIIA using  $r_{\text{C}}$ . These points are not surprising given that the  $S_6 \rightarrow C_3$  distortion involves, for example, the subtle deformation of the  $\text{PF}_6^-$  coordination sphere, as will be discussed below.

We now examine the ring center wells in greater detail. The  $S_6 \rightarrow C_3$  distortion makes the ring center wells in the northern hemisphere deeper and more varied in depth. The  $C_6(1)$  well at the north pole and its adjacent  $C_6(2)$  wells thus formed are the deepest and the second deepest, respectively, among all the wells. The deepest  $C_6(1)$  well is in the vicinity of the experimental high-occupancy disordered site at  $r = 1.436$  Å (*i.e.*, the orange point in Fig. 1). Although we have above focused on the potential energies at  $r = 1.375$  Å underneath the  $C_6(1)$  ring (*i.e.*,  $V_{\text{es+exrep+pol+disp}}^{6\text{K}} = -10.68$  kcal mol<sup>-1</sup> and  $\Delta E_{\text{RI-MP2}} = -10.72$  kcal mol<sup>-1</sup>, as shown without parentheses in Table 1), the energies become slightly more stable as  $\text{Li}^+$  further approaches the high-occupancy disordered site; indeed,  $V_{\text{es+exrep+pol+disp}}^{6\text{K}} = -10.72$  kcal mol<sup>-1</sup> and  $\Delta E_{\text{RI-MP2}} = -10.73$  kcal mol<sup>-1</sup> at  $r = 1.4375$  Å, as shown in parentheses in Table 1. These results illustrate that our model is reasonably accurate, and the interactions beyond the next nearest neighbors are not indispensable to the formation of the deepest well near the high-occupancy disordered site. In the southern hemisphere, the  $C_6(1')$  well at the south pole is the deepest (Table 1). It is as deep as most of the ring center wells in the northern hemisphere but noticeably shallow compared to its northern counterpart, the  $C_6(1)$  well, and the  $C_6(2)$  wells. The  $C_6(1')$  well is close to the experimental low-occupancy disordered site at  $r = 1.366$  Å (*i.e.*, the maroon point in Fig. 1a). The small but nonzero occupancy of  $\text{Li}^+$  there ( $\approx 0.24$ )<sup>31</sup> might suggest a slight underestimation of the depth of the  $C_6(1')$  well in our calculation. The  $C_5(1)$  wells near the north pole are more stable than many of the  $C_6$  wells (Table 1), which highlights the importance of the  $S_6 \rightarrow C_3$  distortion and the  $\text{PF}_6^-$  coordination. Recall here that any  $C_5$  well is less stable than the  $C_6$  wells in the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (40 K) model<sup>25</sup> and the isolated  $\text{Li}^+@C_{60}^-$  model.<sup>38</sup>



### C. Mechanism of the inversion-asymmetric well deepening and shallowing

To analyze how the  $S_6 \rightarrow C_3$  distortion and the intermolecular interactions deepen or diminish the ring center wells, we performed EDA calculations for the wells and for the  $C_5(2)$  and  $C_5(2')$  sites, at which no wells are located, in the  $[Li^+@C_{60}]6PF_6^-$  (6 K) model. See Section IIC for the details of the EDA method. The  $\Delta E_{RI-MP2}$  energy of every well/site was decomposed into four EDA components with different types of interactions (*i.e.*,  $\Delta E_{es}$ ,  $\Delta E_{exrep}$ ,  $\Delta E_{pol}$ , and  $\Delta E_{disp}$  in Table 1 and Tables S8, S9, ESI†). The subsequent subtraction of each EDA component for a southern well/site (*e.g.*, the  $C_6(1')$  well) from its corresponding one for the northern counterpart (*e.g.*, the  $C_6(1)$  well) yielded the bar chart in Fig. 4. In other words, the bar chart shows the decomposition of the energy gap between every pair of northern and southern wells/sites (*i.e.*, the gap indicated by every RI-MP2 bar) into the EDA components (*i.e.*, the gaps indicated by the remaining bars), thereby helping reveal the origins of the inversion-asymmetric well deepening and shallowing.

The relative energy stabilization (destabilization) in the northern (southern) hemisphere makes all the RI-MP2 bars (blue, Fig. 4) have negative values, except for the negligible bar for the  $C_5(2)$  and  $C_5(2')$  pair. Because the degree of (de)stabilization is, on the whole, greater near the poles than near the equator, the RI-MP2 bars for pairs of near-pole wells have larger absolute values. The same features of energy gaps, namely negative values for any pairs of wells and large absolute values for near-pole pairs, are seen in the es bars (light blue, Fig. 4), and the es bars are the most dominant components of all. Further fragment-pair decomposition for the es bars (Fig. S10a, ESI†) reveals that the differences both in the  $Li^+ \cdots C_{60}$  electrostatic energy and in the  $Li^+ \cdots 6PF_6^-$  one make major

contributions, giving rise to the aforementioned features. Because of the  $C_{60}$  cage distortion,  $Li^+$  suffers weaker  $Li^+ \cdots C_{60}$  electrostatic repulsion at a northern well than at its southern counterpart (Fig. S10a, ESI†). Besides, the  $S_6 \rightarrow C_3$  distortion involves inward (outward) displacement of the three  $PF_6^-$  anions near the northern (southern) hemisphere of  $C_{60}$ ; in fact, the electronegative F atoms near the northern hemisphere (Fig. 1b) and those near the southern one show radial shifts,  $\Delta r_F = r_F^{6K} - r_F^{40K}$ , of  $-0.013$  and  $0.003$  Å, respectively. This displacement strengthens (weakens) the  $Li^+ \cdots 6PF_6^-$  electrostatic attraction when  $Li^+$  is at a well near the north (south) pole (Fig. S10a and Tables S8, S9, ESI†). The  $Li^+ \cdots 6PF_6^-$  electrostatic attraction and its distortion-induced change are not negligible due to their long-range nature, which is one reason the well deepening and shallowing (Fig. 3b) are not directly associated with the  $r_C$  changes (Fig. S4, ESI†). In contrast, exchange repulsion has a short-range nature. As a result, the differences in the  $Li^+ \cdots C_{60}$  exchange repulsion energy are dominant, and those in the  $Li^+ \cdots 6PF_6^-$  one are negligible (Fig. S10b, ESI†). Thus, the exrep bars (light green, Fig. 4) evidently reflect the  $r_C$  changes. The exrep bars for three near-pole pairs, namely  $C_6(1)$  and  $C_6(1')$ ,  $C_6(2)$  and  $C_6(2')$ , and  $C_5(1)$  and  $C_5(1')$ , have negative values, as with the RI-MP2 bars and the es ones. That is because when  $Li^+$  is at a well near the north (south) pole, the outward (inward) displacement of nearby C atoms makes  $Li^+$  suffer weaker (stronger) exchange repulsion with them (see Table 1 for the mean distances,  $d_{Li-C}$ , and Tables S8 and S9 (ESI†) for the  $Li^+ \cdots C_{60}$  exchange repulsion energies). Unlike the exrep bars for the near-pole pairs, those for the near-equator pairs, namely  $C_6(3)$  and  $C_6(3')$ ,  $C_6(4)$  and  $C_6(4')$ , and  $C_5(2)$  and  $C_5(2')$ , have positive values. This is because the  $\Delta r_C$  values of near-equator C atoms have features opposite in sign to those of near-pole ones (Fig. S4, ESI†), as discussed above; that is, the C atoms right to the north (south) of the equator show the inward (outward) displacement. The pol bars (light pink, Fig. 4) have positive values for any pairs of wells, and the values are large particularly for near-pole pairs. The fragment-pair decomposition indicates that the differences in the  $Li^+ \cdots C_{60}$  polarization energy and/or the many-body one make major contributions (Fig. S10c, ESI†). The differences in the  $Li^+ \cdots C_{60}$  polarization energy have features opposite in sign to those in the  $Li^+ \cdots C_{60}$  exchange repulsion energy. In fact, the green bars in Fig. S10c (ESI†), which indicate differences in the  $Li^+ \cdots C_{60}$  polarization energy, have positive (negative) values for the near-pole (near-equator) pairs. This is because long (short) distances between  $Li^+$  and the nearby C atoms lead to the formation of weak (strong) polarization attraction (Tables S8 and S9, ESI†). The disp bars (wine red, Fig. 4) show that the differences in dispersion energy make quite minor contributions. In summary, the northern well deepening (southern well shallowing), particularly near the pole, is ascribed to the decreases (increases) in electrostatic and exchange repulsion energies, which override the concurrent increase (decrease) in polarization energy (Fig. 4).<sup>64</sup> The changes in these three energies are attributed to the  $S_6 \rightarrow C_3$  symmetry lowering, or the inversion-symmetry breaking, of the local structure; that is,

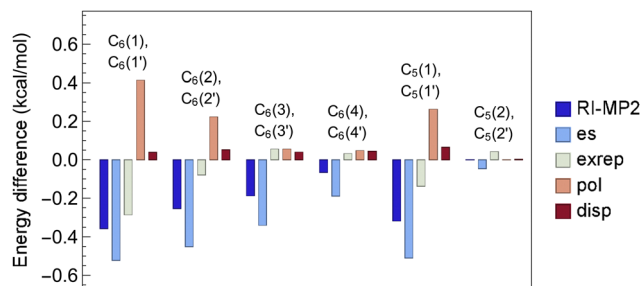


Fig. 4 Energy decomposition of the potential energy difference between the point  $(x,y,z)$  underneath a northern ring center (*e.g.*, the  $C_6(1)$  center) and the point  $(x',y',z')$  underneath its southern counterpart (*e.g.*, the  $C_6(1')$  center) in the  $[Li^+@C_{60}]6PF_6^-$  (6 K) model. The decomposition is expressed as  $E_{RI-MP2}(x,y,z) - E_{RI-MP2}(x',y',z') = \sum_{\alpha}^{es,exrep,pol,disp} \{E_{\alpha}(x,y,z) - E_{\alpha}(x',y',z')\}$ . Using the energies listed in Table 1, this can be rewritten as  $\Delta E_{RI-MP2}(x,y,z) - \Delta E_{RI-MP2}(x',y',z') = \sum_{\alpha}^{es,exrep,pol,disp} \{\Delta E_{\alpha}(x,y,z) - \Delta E_{\alpha}(x',y',z')\}$ . The two points,  $(x,y,z)$  and  $(x',y',z')$ , are 1.375 Å apart from the  $C_{60}$  cage center, and if the  $S_6 \rightarrow C_3$  symmetry lowering were absent, they would be equivalent; in other words, all energy bars in the figure would disappear. In the graph legends, only the subscript of every energy difference (*e.g.*,  $\alpha$ , which denotes the interaction type) is shown for brevity.

the C<sub>60</sub> cage distortion has striking effects on any of the three, whilst the deformation of the PF<sub>6</sub><sup>−</sup> coordination sphere partly accounts for the electrostatic energy change (Fig. S10, ESI†).

#### D. Distortion-induced localization of nuclear wave functions of Li<sup>+</sup>

For the [Li<sup>+</sup>@C<sub>60</sub>]6PF<sub>6</sub><sup>−</sup> (40 K) model, we previously examined the eight lowest-energy nuclear wave functions of Li<sup>+</sup>,<sup>25</sup> which are herein designated as  $\psi_k^{40K}$  ( $k = 1, 2, \dots, 8$ ) in the order of increasing energy (Table 2). The ground state is nearly twofold degenerate, and the relevant wave functions belong to different irreducible representations (*i.e.*,  $\psi_1^{40K}$ , labeled A<sub>g</sub>, and  $\psi_2^{40K}$ , labeled A<sub>u</sub>).<sup>25</sup> They compose a gerade–ungerade pair and can be expressed as

$$\psi_1^{40K} = \frac{1}{\sqrt{2}}(\psi_{\text{north}}^{\text{bs}} + \psi_{\text{south}}^{\text{bs}}), \quad (6)$$

$$\psi_2^{40K} = \frac{1}{\sqrt{2}}(\psi_{\text{north}}^{\text{bs}} - \psi_{\text{south}}^{\text{bs}}). \quad (7)$$

Here,  $\psi_{\text{north}}^{\text{bs}}$  and  $\psi_{\text{south}}^{\text{bs}}$  are orthonormal broken-symmetry wave functions essentially centered on each of the northern and southern C<sub>6</sub>(1) wells, while just slightly delocalized over the nearby C<sub>6</sub>(2) wells. The symmetric and antisymmetric superpositions in eqn (6) and (7) make  $\psi_1^{40K}$  and  $\psi_2^{40K}$  evenly localized at the two C<sub>6</sub>(1) wells, near the disordered sites at 40 K, but give rise to only an extremely small energy splitting of  $2.21 \times 10^{-8}$  kcal mol<sup>−1</sup>.<sup>25</sup> This is partly because of the long distance between the wells and the near-equator barriers, which are formed by the cage elongation along the S<sub>6</sub> axis. Similarly, the excited-state  $\psi_3^{40K}$  and  $\psi_4^{40K}$  wave functions, labeled E<sub>u</sub>, and the excited-state  $\psi_5^{40K}$  and  $\psi_6^{40K}$  ones, labeled E<sub>g</sub>, are nearly degenerate, with a splitting of  $1.66 \times 10^{-5}$  kcal mol<sup>−1</sup> (Table 2). The four wave functions are considerably localized at the C<sub>6</sub>(2) wells near the two poles, while moderately delocalized over the C<sub>6</sub>(1) wells. They compose another gerade–ungerade pair, which arises from superpositions of four broken-symmetry wave functions. Two of the broken-symmetry wave functions and the remaining two are mostly localized at the northern C<sub>6</sub>(2) wells and the southern C<sub>6</sub>(2) ones, respectively. The excited-state

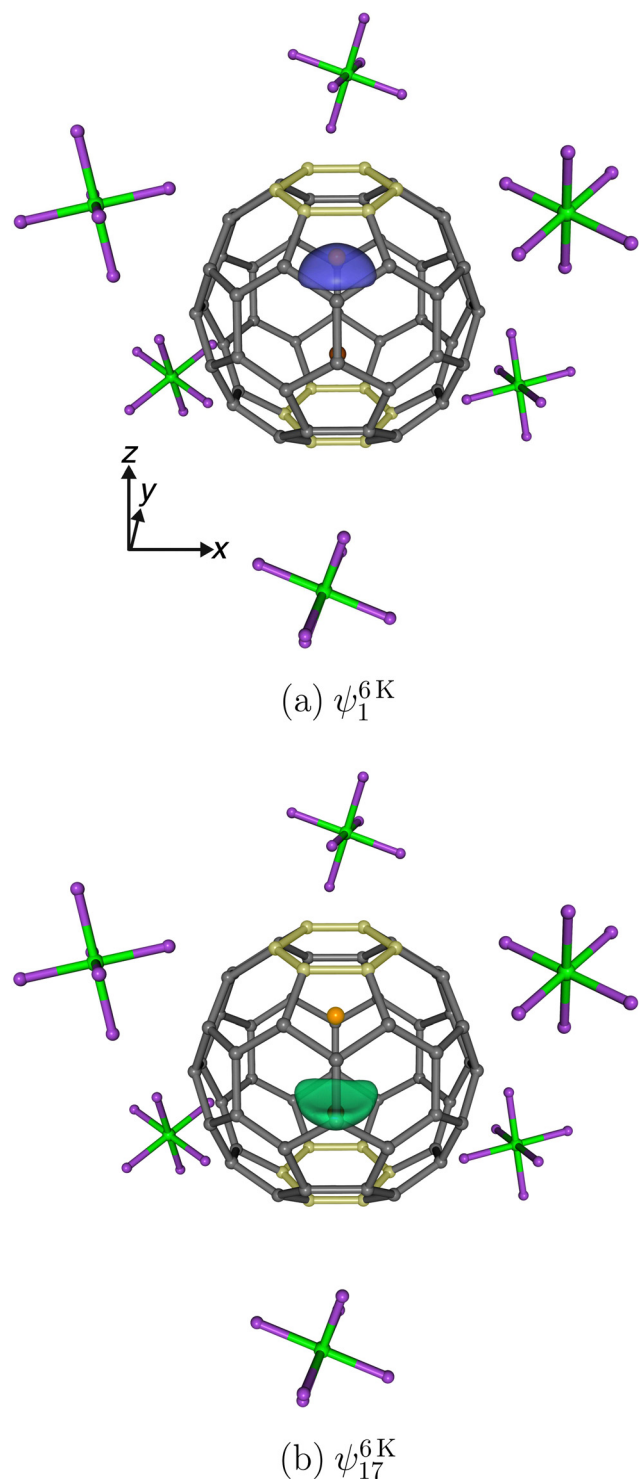
$\psi_7^{40K}$  and  $\psi_8^{40K}$  wave functions, labeled A<sub>g</sub> and A<sub>u</sub>, respectively, are significantly localized at the C<sub>6</sub>(2) wells and moderately localized at the C<sub>6</sub>(1) wells. They compose a gerade–ungerade pair with a splitting of  $1.05 \times 10^{-4}$  kcal mol<sup>−1</sup>. Overall, a gerade–ungerade pair found in a high-energy region has large energy splitting, to which symmetry-adapted superposition has great physical relevance.

To reveal how the inversion-symmetry breaking of the local structure deforms nuclear wave functions of Li<sup>+</sup>, we examined the superpositions induced by the perturbation of  $V_{\text{es+exrep+pol+disp}}^{6K} - V_{\text{es+exrep+pol+disp}}^{40K}$  using the overlap matrix *S* between the set of  $\psi_k^{40K}$  ( $k = 1, 2, \dots, 801$ ) and that of  $\psi_l^{6K}$  ( $l = 1, 2, \dots, 802$ ). Here,  $\{\psi_k^{6K}\}$  is the set of the nuclear wave functions of Li<sup>+</sup> in the [Li<sup>+</sup>@C<sub>60</sub>]6PF<sub>6</sub><sup>−</sup> (6 K) model, which, as well as  $\{\psi_k^{40K}\}$ , is arranged in the order of increasing energy. The elements of the  $200 \times 200$  and the  $31 \times 31$  submatrix of *S* (*i.e.*, low-energy blocks) and the squared norm of each column of them are visually represented in Fig. S11 and S12 (ESI†), respectively. The diagonal band structure of the  $200 \times 200$  submatrix (Fig. S11a, ESI†) and the large norms of most of its columns (Fig. S12b, ESI†) indicate that the perturbation induces superpositions only among low-energy unperturbed states  $\{\psi_k^{40K}\}$ , particularly those close in energy, to yield a low-energy perturbed state,  $\psi_l^{6K}$ . Such superpositions involve gerade–ungerade mixing (Fig. S11b, ESI†), namely A<sub>g</sub>–A<sub>u</sub> or E<sub>g</sub>–E<sub>u</sub> mixing, so that many of the perturbed wave functions  $\{\psi_l^{6K}\}$ , labeled A or E, have spatial distributions dramatically different from the unperturbed ones  $\{\psi_k^{40K}\}$ . For instance, the eight lowest-energy unperturbed wave functions,  $\psi_k^{40K}$  ( $k = 1, 2, \dots, 8$ ), are evenly distributed in the northern and southern hemispheres to satisfy the inversion symmetry, whereas the gerade–ungerade mixing among them (Fig. S11b, ESI†) yields the four lowest-energy perturbed ones,  $\psi_l^{6K}$  ( $l = 1, 2, \dots, 4$ ), localized in the northern hemisphere, and four higher-energy ones,  $\psi_l^{6K}$  ( $l = 17, 28, 29, 31$ ), mostly localized in the southern hemisphere, as will be discussed below. This is clearly associated with the northern well deepening and the southern well shallowing. The deepening and shallowing make the low-lying energy levels considerably more discrete in the [Li<sup>+</sup>@C<sub>60</sub>]6PF<sub>6</sub><sup>−</sup> (6 K) model (Fig. S13b, ESI†) than in the [Li<sup>+</sup>@C<sub>60</sub>]6PF<sub>6</sub><sup>−</sup> (40 K) model (Fig. S13a, ESI†). In fact, in the frequency range up to 5 THz,

**Table 2** Irreducible representations (irrep.), energies (kcal mol<sup>−1</sup>), and frequencies (THz) for the eight lowest-energy nuclear wave functions of Li<sup>+</sup> in each of the local structure models. Every frequency corresponds to the transition from the nuclear ground state,  $\psi_1^{40K}$  or  $\psi_1^{6K}$ , to the relevant state. For the [Li<sup>+</sup>@C<sub>60</sub>]6PF<sub>6</sub><sup>−</sup> (6 K) model, the data for the 17th-lowest energy wave function,  $\psi_{17}^{6K}$ , which is mostly localized at the low-occupancy disordered site (Fig. 5b and 6f), are also listed

| [Li <sup>+</sup> @C <sub>60</sub> ]6PF <sub>6</sub> <sup>−</sup> (40 K) <sup>a</sup> |                |        |           | [Li <sup>+</sup> @C <sub>60</sub> ]6PF <sub>6</sub> <sup>−</sup> (6 K) |        |        |           |
|--------------------------------------------------------------------------------------|----------------|--------|-----------|------------------------------------------------------------------------|--------|--------|-----------|
| State                                                                                | Irrep.         | Energy | Frequency | State                                                                  | Irrep. | Energy | Frequency |
| $\psi_1^{40K}$                                                                       | A <sub>g</sub> | −9.81  | —         | $\psi_1^{6K}$                                                          | A      | −10.04 | —         |
| $\psi_2^{40K}$                                                                       | A <sub>u</sub> | −9.81  | 0.00      | $\{\psi_2^{6K}, \psi_3^{6K}\}$                                         | E      | −9.90  | 1.49      |
| $\{\psi_3^{40K}, \psi_4^{40K}\}$                                                     | E <sub>u</sub> | −9.72  | 1.02      | $\psi_4^{6K}$                                                          | A      | −9.86  | 1.85      |
| $\{\psi_5^{40K}, \psi_6^{40K}\}$                                                     | E <sub>g</sub> | −9.72  | 1.02      | $\{\psi_5^{6K}, \psi_6^{6K}\}$                                         | E      | −9.82  | 2.33      |
| $\psi_7^{40K}$                                                                       | A <sub>g</sub> | −9.70  | 1.19      | $\{\psi_7^{6K}, \psi_8^{6K}\}$                                         | E      | −9.76  | 2.91      |
| $\psi_8^{40K}$                                                                       | A <sub>u</sub> | −9.70  | 1.19      | $\psi_{17}^{6K}$                                                       | A      | −9.67  | 3.87      |

<sup>a</sup> The eight wave functions in the [Li<sup>+</sup>@C<sub>60</sub>]6PF<sub>6</sub><sup>−</sup> (40 K) model,  $\psi_k^{40K}$  ( $k = 1, 2, \dots, 8$ ), have recently been examined in detail.<sup>25</sup>



**Fig. 5** Isosurface plots (side view) of two nuclear wave functions of  $\text{Li}^+$  in the  $[\text{Li}^+@C_{60}]\text{6PF}_6^-$  (6 K) model: (a) the ground-state  $\psi_1^{6K}$  wave function, strongly localized at the high-occupancy disordered site (i.e., the orange point), and (b) the excited-state  $\psi_{17}^{6K}$  wave function, strongly localized at the low-occupancy disordered site (i.e., the maroon point). The blue and green isosurfaces were generated using the isovalues of  $\pm 0.02 \text{ Bohr}^{-3/2}$ . The  $C_6(1)$  ring and the  $C_6(1')$  ring are indicated by yellow highlights.

or in the energy range of the ground-state energy to about  $0.48 \text{ kcal mol}^{-1}$  above it, there are only 30 states in the

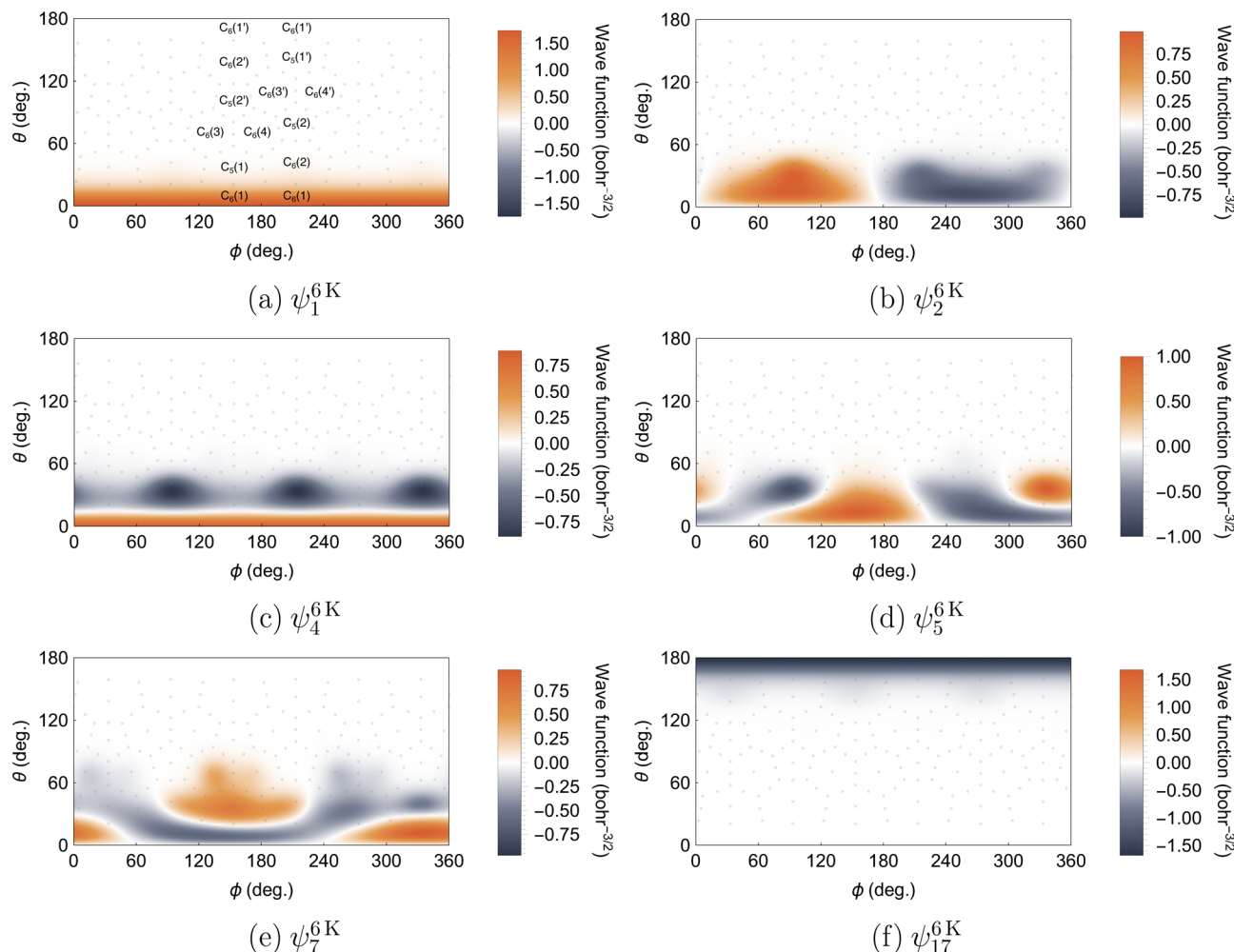
$[\text{Li}^+@C_{60}]\text{6PF}_6^-$  (6 K) model, while there are 80 states in the  $[\text{Li}^+@C_{60}]\text{6PF}_6^-$  (40 K) model.

Focusing on several low-energy nuclear wave functions of  $\text{Li}^+$  in the  $[\text{Li}^+@C_{60}]\text{6PF}_6^-$  (6 K) model,  $\psi_l^{6K}$  ( $l = 1, 2, \dots, 8, 17, 28, 29, 31$ ), we discuss the superpositions, spatial distributions, and energies in greater detail. Fig. 5 and Fig. S14 (ESI†) show the isosurface plots of the wave functions, and Fig. 6 and Fig. S15 (ESI†) show the polar plots of them. The ground state is nondegenerate (Table 2), unlike that in the  $[\text{Li}^+@C_{60}]\text{6PF}_6^-$  (40 K) model, and its wave function,  $\psi_1^{6K}$ , belongs to the irreducible representation A. The ground-state energy level,  $-10.04 \text{ kcal mol}^{-1}$ , is well separated from the low-lying excited-state ones listed in Table 2. Similar to  $\psi_{\text{north}}^{\text{bs}}$ ,  $\psi_1^{6K}$  is essentially localized at the deepest  $C_6(1)$  well, near the high-occupancy disordered site at 6 K,<sup>31</sup> and just slightly delocalized over the second-deepest  $C_6(2)$  wells (Fig. 5a and 6a). A wave function similar to  $\psi_{\text{south}}^{\text{bs}}$ ,  $\psi_{17}^{6K}$ , is found to have a fairly high energy of  $-9.67 \text{ kcal mol}^{-1}$  (3.87 THz, Table 2). The excited-state  $\psi_{17}^{6K}$  wave function is localized at the  $C_6(1')$  well, near the low-occupancy disordered site at 6 K,<sup>31</sup> and just slightly delocalized over the  $C_6(2')$  wells (Fig. 5b and 6f). A numerical analysis of  $S$  (Fig. S11b, ESI†) confirmed that  $\psi_1^{6K}$  and  $\psi_{17}^{6K}$  can indeed be approximately expressed as

$$\psi_1^{6K} \approx \frac{1}{\sqrt{2}}(\psi_1^{40K} + \psi_2^{40K}) = \psi_{\text{north}}^{\text{bs}}, \quad (8)$$

$$\psi_{17}^{6K} \approx -\frac{1}{\sqrt{2}}(\psi_1^{40K} - \psi_2^{40K}) = -\psi_{\text{south}}^{\text{bs}}. \quad (9)$$

These results indicate that the  $A_g$ – $A_u$  mixing in eqn (8) and (9), induced by the inversion-symmetry breaking, makes  $\psi_1^{6K}$  localized at the northern  $C_6(1)$  well and  $\psi_{17}^{6K}$  localized at the southern  $C_6(1')$  well. In addition, the mixing splits the nearly degenerate energy level of  $\psi_1^{40K}$  and  $\psi_2^{40K}$  into the ground-state level of  $\psi_1^{6K}$  and the excited-state one of  $\psi_{17}^{6K}$  (Table 2). Given the near-degeneracy of the gerade–ungerade pair of  $\psi_1^{40K}$  and  $\psi_2^{40K}$ , there is an alternative but essentially equivalent interpretation of this distortion-induced energy splitting, without any need for gerade–ungerade mixing.<sup>65</sup> That is, the broken-symmetry states of  $\psi_{\text{north}}^{\text{bs}}$  and  $\psi_{\text{south}}^{\text{bs}}$  can be thought of as approximate eigenstates both in the  $[\text{Li}^+@C_{60}]\text{6PF}_6^-$  (40 K) model and in the  $[\text{Li}^+@C_{60}]\text{6PF}_6^-$  (6 K) one. The inversion-symmetry breaking of the local structure is accompanied by the northern well deepening and the southern well shallowing, thus stabilizing  $\psi_{\text{north}}^{\text{bs}}$  and destabilizing  $\psi_{\text{south}}^{\text{bs}}$  in the  $[\text{Li}^+@C_{60}]\text{6PF}_6^-$  (6 K) model.<sup>65</sup> Supposing that a sudden distortion-induced perturbation of  $V_{\text{es+exrep+pol+disp}}^{6K} - V_{\text{es+exrep+pol+disp}}^{40K}$  is applied to the initial state of  $\psi_{\text{south}}^{\text{bs}}$ , which is localized around the southern  $C_6(1')$  well, the coherent transfer to the northern  $C_6(1)$  well must be slow. This hypothetical case implies that the coupling of the motion of  $\text{Li}^+$  to the motions of the surrounding  $C_{60}$  cage, in part constituting the phonon reservoir, would be important to understand the dynamic aspect of the two-site disorder–order transition. This aspect is beyond the scope of this study. As shown in Fig. S11b (ESI†),  $E_g$ – $E_u$  mixing primarily among nearly



**Fig. 6** Polar plots of six nuclear wave functions of  $\text{Li}^+$  in the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (6 K) model, which are selected from the 17 lowest-energy ones and presented in the order of increasing energy. The radial distance  $r$  is 1.375 Å. Panels (a), (c), and (f) show the ground-state  $\psi_1^{6K}$  wave function, the excited-state  $\psi_4^{6K}$  one, and the excited-state  $\psi_7^{6K}$  one, respectively, all of which belong to the irreducible representation A. Each of the remaining three excited-state ones, namely (b)  $\psi_2^{6K}$ , (d)  $\psi_5^{6K}$ , and (e)  $\psi_7^{6K}$ , forms a basis for the degenerate irreducible representation E, together with  $\psi_3^{6K}$ ,  $\psi_6^{6K}$ , and  $\psi_8^{6K}$  (Table 2 and Fig. S14, S15, ESI†), respectively. The north pole and the south pole are spread out over all azimuth angles  $\phi$  at the bottom ( $\theta = 0^\circ$ ) and the top ( $\theta = 180^\circ$ ) of each map, respectively. The positions underneath the C atoms, bond centers, and ring centers, except those underneath the  $C_{60}(1)$  and  $C_{60}(1')$  centers, are indicated by gray points. The plot range, shown in each bar legend, was determined using the largest amplitude on the shell with  $r = 1.375$  Å.

degenerate  $\psi_k^{40K}$  ( $k = 3, 4, \dots, 6$ ) yields two stable states,

$$\psi_2^{6K} \approx -\frac{1}{2}(\psi_3^{40K} + \psi_4^{40K} - \psi_5^{40K} + \psi_6^{40K}), \quad (10)$$

$$\psi_3^{6K} \approx -\frac{1}{2}(-\psi_3^{40K} + \psi_4^{40K} + \psi_5^{40K} + \psi_6^{40K}), \quad (11)$$

which are largely localized at the  $C_{60}(2)$  wells and moderately delocalized over the  $C_{60}(1)$  well in the northern hemisphere (Fig. 6b and Fig. S14b, c, S15a, ESI†), and their unstable southern counterparts (Fig. S14j, k and S15d, e, ESI†),

$$\psi_{28}^{6K} \approx \frac{1}{2}(\psi_3^{40K} - \psi_4^{40K} + \psi_5^{40K} + \psi_6^{40K}), \quad (12)$$

$$\psi_{29}^{6K} \approx -\frac{1}{2}(\psi_3^{40K} + \psi_4^{40K} + \psi_5^{40K} - \psi_6^{40K}). \quad (13)$$

The former two have energies of  $-9.90 \text{ kcal mol}^{-1}$  (1.49 THz,

Table 2), and the latter two have energies of  $-9.57 \text{ kcal mol}^{-1}$  (4.87 THz). Similarly,  $A_g-A_u$  mixing primarily between nearly degenerate  $\psi_7^{40K}$  and  $\psi_8^{40K}$  yields a stable state,

$$\psi_4^{6K} \approx \frac{1}{\sqrt{2}}(\psi_7^{40K} - \psi_8^{40K}), \quad (14)$$

which is significantly localized at the  $C_{60}(2)$  wells and moderately localized at the  $C_{60}(1)$  well in the northern hemisphere (Fig. 6c and Fig. S14d, ESI†), and its unstable southern counterpart (Fig. S14l and S15f, ESI†),

$$\psi_{31}^{6K} \approx \frac{1}{\sqrt{2}}(\psi_7^{40K} + \psi_8^{40K}). \quad (15)$$

The former has an energy of  $-9.86 \text{ kcal mol}^{-1}$  (1.85 THz, Table 2), and the latter has an energy of  $-9.56 \text{ kcal mol}^{-1}$  (5.02 THz). The fifth- to eighth-lowest energy states in the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (6 K) model,  $\psi_l^{6K}$  ( $l = 5, 6, \dots, 8$ ), arise from



$E_g$ – $E_u$  mixing (Fig. S11b, ESI<sup>†</sup>), particularly among higher-energy unperturbed states of  $\psi_k^{40K}$  ( $k \geq 9$ ). Their wave functions (Fig. 6d, e and Fig. S14e–h, S15b, c, ESI<sup>†</sup>), as well as all the other wave functions below  $-9.67$  kcal mol<sup>-1</sup> (*i.e.*, the energy of  $\psi_{17}^{6K} \approx -\psi_{south}^{bs}$ ), have a large amplitude in the northern hemisphere. The northern well deepening makes many low-energy wave functions strongly trapped in the northern hemisphere *via* gerade–ungerade mixing,<sup>65</sup> leading to the discretization of low-lying energy levels in the  $[Li^+@C_{60}]6PF_6^-$  (6 K) model.

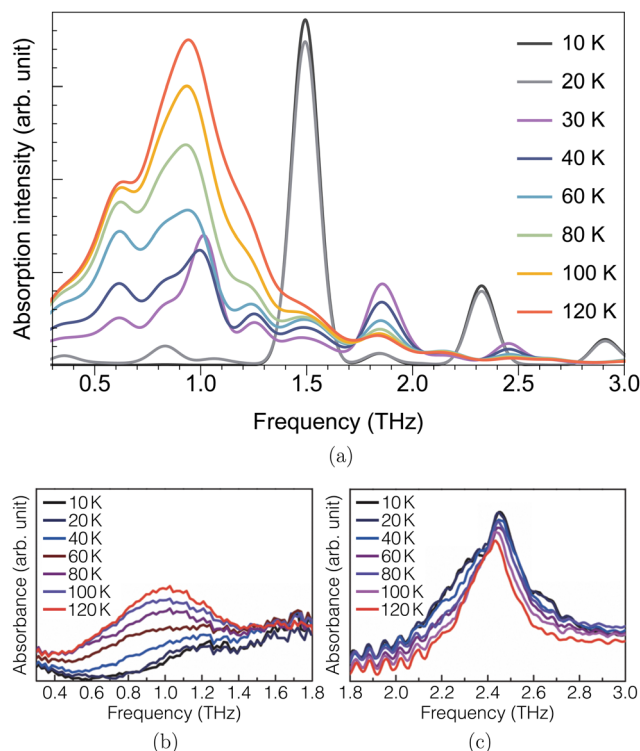
### E. Effects of the wave function localization on the THz absorption spectra

Assuming that the nuclear motion of  $Li^+$  is decoupled from the motions of  $C_{60}$  and  $PF_6^-$ , we investigate the contribution of the  $Li^+$  motion to the THz absorption spectra of the salt. In temperature-dependent THz absorption spectroscopy experiments, Suzuki *et al.* pressed a crystalline salt to form a pellet and, for low-temperature measurements, further mixed it with pristine  $C_{60}$  crystals.<sup>32</sup> Given that the  $[Li^+@C_{60}]6PF_6^-$  centers are randomly oriented, a stick absorption spectrum can be written as

$$\sigma_{abs}(\omega) = \frac{4\pi^2\omega}{3c} \sum_i p_i \sum_f |\langle \psi_i | \mu | \psi_f \rangle|^2 \delta(E_f - E_i - \hbar\omega), \quad (16)$$

where  $E_i$  and  $E_f$  are the energy of an initial nuclear state of  $Li^+$ ,  $\psi_i$ , and that of a final one,  $\psi_f$ , respectively, and  $p_i$  is the Boltzmann population of the initial state.<sup>38,66,67</sup> The eigensolutions of the  $[Li^+@C_{60}]6PF_6^-$  (6 K) model and those of the  $[Li^+@C_{60}]6PF_6^-$  (40 K) model, discussed in Section IIID, were employed to obtain the spectra below and above  $T_C$ , respectively. The  $\mu$  in eqn (16), the electronic expectation value of the dipole moment, was replaced by the position vector of  $Li^+$ ,  $\mathbf{x} = (x, y, z)$ , in actual calculations. This approximation, previously applied to  $Li^+@C_{60}^-$ ,<sup>38</sup> was numerically justified for both of our  $[Li^+@C_{60}]6PF_6^-$  models (see Fig. S16 and S17 and the explanation on p. 28 in ESI<sup>†</sup>). Every stick spectrum was convoluted with a Gaussian function with a full width at half maximum (FWHM) of  $5$  cm<sup>-1</sup> ( $\approx 0.15$  THz).

The convoluted spectra and the stick spectra in the frequency range of 0.3 to 3.0 THz are shown in Fig. 7a and Fig. S18, S19 (ESI<sup>†</sup>), respectively. In addition, in Fig. S18 (ESI<sup>†</sup>), the spectral contributions of the transitions from the ground state (*i.e.*,  $\psi_1^{40K}$  and  $\psi_2^{40K}$  for the spectra above  $T_C$  and  $\psi_1^{6K}$  for those below  $T_C$ ) to excited states are separated from those of the transitions among the excited states. At 120 K, there is a prominent, broad peak at around 0.94 THz (red line, Fig. 7a). This peak arises mainly from transitions among excited states (Fig. S18a, ESI<sup>†</sup>). Each of the transitions has only a weak intensity because of the small population of the initial excited-state level, but the group of them with different frequencies yields the strong broad peak (Fig. S18a and S19a, ESI<sup>†</sup>). As the temperature decreases to 60 K, the peak intensity decreases substantially because of the depopulation of the excited-state levels upon cooling (cyan line, Fig. 7a). The depopulation is accompanied by the population increase at



**Fig. 7** Temperature-dependent THz absorption spectra of the  $[Li^+@C_{60}]6PF_6^-$  salt. Panel (a) shows the theoretical spectra below  $T_C$  (grayscale lines) and above  $T_C$  (chromatic lines). Only the contribution of the  $Li^+$  motion to the spectra is considered in the calculations. Panels (b) and (c) show the experimental spectra in the frequency ranges of 0.3 to 1.8 THz and 1.8 to 3.0 THz, respectively. To obtain the spectra in the two panels, Suzuki *et al.* used different spectrometers, namely a THz time-domain spectrometer for panel (b) and a Fourier-transform far-infrared spectrometer for panel (c). Adapted from ref. 32 with permission from the PCCP Owner Societies. Note that the color scheme is not consistent across the panels.

the ground-state level. Although not evident in the spectrum at 60 K owing to the spectral broadness, a peak associated with the ground state actually emerges (see the shaded peak and the relatively strong stick intensities at 1.02 THz in Fig. S18d and S19d, ESI<sup>†</sup>). The hidden peak at 1.02 THz corresponds to the transitions from the nearly degenerate ground state to the four lowest-energy excited states, namely the  $\psi_1^{40K}(A_g) \rightarrow \{\psi_3^{40K}, \psi_4^{40K}\}(E_u)$  and  $\psi_2^{40K}(A_u) \rightarrow \{\psi_5^{40K}, \psi_6^{40K}\}(E_g)$  transitions (Table 2). To conserve the inversion symmetry, these excitations take place both around the north pole and around the south pole. The weak peak at 1.85 THz (cyan line, Fig. 7a) also arises largely from transitions from the ground state (Fig. S18d, ESI<sup>†</sup>), namely the  $\psi_1^{40K}(A_g) \rightarrow \{\psi_{15}^{40K}, \psi_{16}^{40K}\}(E_u)$  transition and the  $\psi_2^{40K}(A_u) \rightarrow \{\psi_{18}^{40K}, \psi_{19}^{40K}\}(E_g)$  and  $\{\psi_{20}^{40K}, \psi_{21}^{40K}\}(E_g)$  transitions. As the temperature decreases to 30 K, right above  $T_C$ , both of the peaks associated with the ground state become more prominent because of its population increase (Fig. 7a and Fig. S18f, S19f, ESI<sup>†</sup>). Meanwhile, the spectral contributions of the transitions among excited states decrease dramatically, particularly below 1 THz (Fig. S18d–f, ESI<sup>†</sup>). Hence, the peak at 1.01 THz at 30 K (purple line, Fig. 7a) is attributed primarily to

the aforementioned transitions from the ground state to the four lowest-energy excited states (Table 2), indicating that the assignments of the peak at around 1 THz vary between 120 and 30 K. In short, the increased (decreased) population of the ground state (excited states), or the localization of  $\text{Li}^+$  at the northern and southern  $\text{C}_6(1)$  wells, upon cooling from 120 to 30 K leads to the significant intensity decrease and the slight blueshift (*i.e.*, the shift from 0.94 to 1.01 THz) of the peak at around 1 THz. Below  $T_C$ , no peaks remain at 1.01 or 1.85 THz, while two new peaks appear at higher frequencies, that is, the strong sharp peak at 1.49 THz and the weak sharp peak at 2.33 THz (black and gray lines, Fig. 7a). Both peaks are attributed to transitions from the ground state (Fig. S18g, h, ESI†), thus evidently reflecting the low-temperature localization of  $\text{Li}^+$  at the northern  $\text{C}_6(1)$  well. The strong peak at 1.49 THz is attributed to the  $\psi_1^{6K}(\text{A}) \rightarrow \{\psi_2^{6K}, \psi_3^{6K}\}(\text{E})$  transition, while the weak peak at 2.33 THz is caused by the  $\psi_1^{6K}(\text{A}) \rightarrow \{\psi_5^{6K}, \psi_6^{6K}\}(\text{E})$  transition (Table 2). The ground state,  $\psi_1^{6K} \approx \psi_{\text{north}}^{\text{bs}}$ , is localized at the northern  $\text{C}_6(1)$  well and the four low-energy excited states involved are also localized near the north pole. This shows that both peaks are attributed to the local excitations around the north pole. The inversion-symmetry breaking and the concurrent northern-well deepening cause the localization of the ground state and the excitations from it, leading to the noticeable blueshifts of the two peaks (*i.e.*, at around 1.01 to 1.49 THz and 1.85 to 2.33 THz) upon cooling below  $T_C$ . We note that the blueshift of each peak involves gerade–ungerade mixing mainly among the states that contribute to the relevant peak; for example, see eqn (8), (10) and (11) for the mixing that causes the peak shift from 1.01 to 1.49 THz and Fig. S11b (ESI†) for the visual representation of the mixing. As the temperature decreases from 20 to 10 K (gray and black lines, respectively, in Fig. 7a), the intensities of both peaks increase somewhat because of the population concentration at the ground-state level (Fig. S19g, h, ESI†).

The overall profiles of the experimental spectra (Fig. 7b and c), except for an intense peak at around 2.45 THz (Fig. 7c) to which the  $\text{Li}^+$  motion does not contribute,<sup>32</sup> are similar to those of the theoretical spectra (Fig. 7a). However, there are several discrepancies, both in the peak assignments and in the interpretation of the temperature dependence of the spectra. Suzuki *et al.* assumed that a rotational band is formed 2.20 THz ( $\approx 0.21 \text{ kcal mol}^{-1}$ ) above a distinct fourfold-degenerate ground-state level,<sup>68</sup> created by the wells at the two disordered sites.<sup>32</sup> Using this intuitive model, they associated the broad peak at around 1 THz (Fig. 7b) with the excited-state rotational band;<sup>32,33</sup> in other words, they considered that the ground state does not contribute to the peak. The intensity decrease of the peak below 120 K (Fig. 7b) was attributed to a thermal population change in the band, namely the suppression of  $\text{Li}^+$  rotations.<sup>32,33</sup> On the other hand, our results indicate that the intensity decrease is also accompanied by the gradual localization of  $\text{Li}^+$  at the two disordered sites. The peak associated with this localization, or the localized ground state, is hidden in the broad peak at 60 K, whereas it becomes dominant at 30 K to yield the modest but still noticeable peak at 1.01 THz (Fig. 7a),

as discussed above. Upon cooling below 120 K, an experimental peak gradually appears at 2.20 THz as a shoulder on the low-frequency tail of the strong peak at around 2.45 THz (Fig. 7c).<sup>32</sup> Further cooling below 30 K causes an additional weak peak at 2.65 THz (*i.e.*, the shoulder on the high-frequency tail in Fig. 7c)<sup>32</sup> and makes the peak at 2.20 THz more prominent (Fig. 7c). Suzuki *et al.* attributed the emergence of the peaks at 2.20 and 2.65 THz to the localization at the two disordered sites above  $T_C$  and that at the high-occupancy disordered site (or the antiferroelectric ordering) below  $T_C$ , respectively.<sup>32,33</sup> However, it has not been settled why the peak at 2.20 THz, attributed to the two-site localization, becomes more prominent below  $T_C$ , at which the subsequent single-site localization proceeds dramatically. As discussed above, our results show that a weak peak appears at around 1.85 THz upon cooling from 120 to 30 K, whereas it disappears and a strong sharp peak appears at 1.49 THz below 20 K (Fig. 7a), which results from the blueshift of the peak at 1.01 THz at 30 K. Therefore, we propose an alternative peak assignment; that is, the theoretical peak at around 1.85 THz, which develops as the two-site localization proceeds upon cooling from 120 K to  $T_C$ , and the theoretical peak at 1.49 THz, which develops as the single-site localization proceeds upon cooling below  $T_C$ , both contribute to the steady intensity increase of the experimental peak at 2.20 THz (Fig. 7c). Although the theoretical peak frequencies are slightly lower than the experimental one, our assignment of the strong peak at 2.20 THz to the two different types of transitions can reasonably explain the steady peak development, particularly below  $T_C$ . It is most likely that the experimental weak peak at 2.65 THz below  $T_C$  (Fig. 7c) corresponds to the theoretical one at 2.33 THz (Fig. 7a), which results from the blueshift of the peak at around 1.85 THz at 30 K. In other words, we attribute the peak appearance at 2.65 THz to the two-site to one-site localization and the resultant blueshift, which is seemingly consistent with the assignment by Suzuki *et al.*<sup>32,33</sup> The mechanisms under consideration are, however, not completely identical. Suzuki *et al.* considered that the antiferroelectric ordering below  $T_C$  causes the peak shift from 2.20 to 2.65 THz.<sup>32,33</sup> Our calculations do not include the collective ordering of  $\text{Li}^+$  ions, but the results demonstrate that the symmetry-breaking-induced single-site localization leads to the peak shift from 1.85 to 2.33 THz. Possible reasons for the peak frequency discrepancies between theory and experiment may include (i) approximations in the adsorbent potential modeling (*e.g.*, the nonperiodic nearest-neighbor truncation, the second-order perturbation treatment of the electron correlation, and the parametric potential fitting) and (ii) the decoupling of the motion of a  $\text{Li}^+$  ion from the motions of its surrounding ions and molecules. The quantum mechanics/molecular mechanics (QM/MM) method,<sup>69–74</sup> particularly its polarizable variants,<sup>75–77</sup> and the ONIOM method<sup>78</sup> may help alleviate the nearest-neighbor truncation problem and clarify the contributions of the antiferroelectric ordering and interactions to the THz spectra. The contributions of the coupled dynamics (*e.g.*, the collective  $\text{Li}^+$  motions and the coupling

of  $\text{Li}^+$  motion with lattice motions) are also of great fundamental interest but beyond the scope of this study.

Zhang *et al.* theoretically predicted that  $\text{Li}^+@C_{60}^-$  shows a strong peak at around 3.75 THz, which was attributed to the first excitation of the twofold-degenerate librational mode in each of the 20  $C_6$  wells.<sup>38</sup> In our theoretical spectra, there is no corresponding peak at any temperature, which is consistent with the experimental results for the  $[\text{Li}^+@C_{60}]\text{PF}_6^-$  salt.<sup>32</sup> This is because the  $C_6$  wells in the salt (Fig. 2) are shallower than those in the electronegative  $C_{60}^-$  cage ( $\approx -15.38 \text{ kcal mol}^{-1}$ ).<sup>38</sup> An experimental sharp peak at around 2.3 to 2.5 THz keeps an extremely strong intensity at any temperature below 300 K.<sup>32</sup> Below 120 K, for instance, the strong peak is at around 2.45 THz (Fig. 7c), as mentioned above. This peak is attributed to the transverse optical phonon mode of the salt.<sup>32,33</sup> It is absent in our theoretical spectra, in which any contributions of the lattice motions are ignored.

## IV. Conclusion

From the point of view of stationary quantum states, we aimed to reveal the molecular origins of the two-site disorder–order transition of the  $[\text{Li}^+@C_{60}]\text{PF}_6^-$  salt. As the temperature decreases below  $T_C = 24 \text{ K}$ , the inversion-symmetry breaking of the local structure of the salt, involving deformations of the  $C_{60}$  cage and the  $\text{PF}_6^-$  coordination sphere, affects intermolecular interactions and, hence, creates deep (shallow) potential wells underneath the northern (southern) ring centers. This inversion-asymmetric perturbation induces gerade–ungerade mixing of the nuclear wave functions of  $\text{Li}^+$ , thereby making low-energy wave functions localized at northern wells and having striking effects on the low-lying energy-level structure (e.g., the higher degree of discreteness). For instance, the twofold near-degeneracy of the ground-state energy level, which appears in the  $[\text{Li}^+@C_{60}]\text{PF}_6^-$  (40 K) model,<sup>25</sup> splits due to  $A_g$ – $A_u$  mixing.<sup>65</sup> The ground-state wave function thus formed is mostly localized at the deepest  $C_6(1)$  well, near the northern high-occupancy disordered site of  $\text{Li}^+$  at 6 K. The distortion-induced off-center localization leads to the formation of a local electric dipole moment and is evident as the blueshifts of the two THz absorption peaks, associated with the ground state, upon cooling below 24 K. Although we do not rule out the existence of the antiferroelectric interactions of the local electric dipole moments in the  $[\text{Li}^+@C_{60}]\text{PF}_6^-$  salt,<sup>31</sup> our results clearly demonstrate the crucial importance of the local structure and interactions in the two-site disorder–order transition of the salt.

The distortion of the host framework we studied is on the order of  $10^{-2} \text{ \AA}$ . Such subtle distortion is often overlooked in materials science. Nonetheless, it can dramatically localize the low-energy nuclear wave functions of endohedral  $\text{Li}^+$  and correlate with macroscopic properties, particularly at low temperatures. The motion of  $\text{Li}^+$  in any high-symmetry host can be extremely sensitive to external symmetry-breaking perturbations or stimuli (e.g., pressures and electric fields) because they

can induce superpositions of the nuclear states and alter the low-lying energy-level structure. This feature may find applications in quantum computing and molecular electronics.<sup>31</sup> The electronic state of a host, particularly if polarizable, can also be sensitive to external stimuli. Our theoretical approach and its extension will help deepen the fundamental understanding of stimuli-responsive host–guest science. Finally, we would like to emphasize the following point. Although absorption peaks in the THz frequency range can, in principle, serve as fingerprints of the endohedral motions of guests and host–guest interactions, the interpretation of the peaks is seldom straightforward, unlike harmonic vibrational peaks in the infrared range. Theoretical THz spectroscopy is of great importance to overcome this difficulty.

## Conflicts of interest

There are no conflicts of interest to declare.

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- 61 Although we did not change any computational methods, the numerical convergence history of the TRLan method was slightly different from that in our previous calculation. As a result, the number of eigensolutions obtained decreased from 803 (in the previous study)<sup>25</sup> to 801 (in this study). We would like to stress that the eigenenergies were numerically the same except for a very few high-lying ones of no interest.
- 62 The relative stabilities of  $\Delta E_{\text{RI-MP2}}$  are reproduced by  $V_{\text{es+exrep+pol+disp}}^{6\text{K}}$  (Table 1), except for the reversed order of the nearly degenerate  $\text{C}_6(3)$  and  $\text{C}_5(1)$  wells.
- 63 A negative (positive) value of  $V_{\text{es+exrep+pol+disp}}^{6\text{K}} - V_{\text{es+exrep+pol+disp}}^{40\text{K}}$  at  $(r, \theta, \phi)$  does not mean that when  $\text{Li}^+$  is fixed at the point in question, the total potential energy of the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (6 K) model is more (less) stable than that of the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (40 K) model. This is because  $V_{\text{es+exrep+pol+disp}}^{6\text{K}}$  and  $V_{\text{es+exrep+pol+disp}}^{40\text{K}}$  are defined relative to different reference geometries; that is,  $V_{\text{es+exrep+pol+disp}}^{6\text{K}}$  and  $V_{\text{es+exrep+pol+disp}}^{40\text{K}}$  are zero (to be precise, nearly zero) when  $\text{Li}^+$  is at the  $\text{C}_{60}$  center of the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (6 K) model and at the  $\text{C}_{60}$  center of the  $[\text{Li}^+@C_{60}]6\text{PF}_6^-$  (40 K) model, respectively.
- 64 This is consistent with our previous finding;<sup>25</sup> that is, the balance between  $\Delta E_{\text{es}}$  and  $\Delta E_{\text{exrep}}$  primarily determines the relative stabilities of the ring center wells (sites) at  $r = 1.375 \text{ \AA}$ .
- 65 As regards the energy splitting of the two lowest-energy states (i.e.,  $\{\psi_1^{40\text{K}}, \psi_2^{40\text{K}}\} \rightarrow \{\psi_1^{6\text{K}}, \psi_{17}^{6\text{K}}\}$ ), there is an interpretation without any need for gerade-ungerade mixing. However, a similar interpretation is less valid for a higher-energy gerade-ungerade pair, since, generally speaking, such a pair has large energy splitting before the inversion-symmetry breaking. The gerade-ungerade mixing is essential and more complicated for higher-energy states, including  $\psi_k^{6\text{K}}$  ( $k = 2, 3, \dots, 16$ ) trapped in the northern hemisphere, as illustrated in Fig. S11b (ESI<sup>†</sup>).
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